

THE TOPOLOGY IS THE SINK: TOPOLOGICAL HALLUCINATION DETECTORS REDUCE TO FIRST-ORDER ATTENTION STATISTICS

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ABSTRACT

Persistent homology (PH) of attention graphs underlies a growing family of hallucination detectors, including the recent TOHA (ACL 2026). We show that, in decoder-only models, these topological features reduce to first-order attention statistics dominated by the *attention sink*, either exactly or up to one computable number. For any symmetric dissimilarity and vertex s , we define the *coning defect* δ_s . Combining classical cone and stability arguments, we show that the entire Vietoris–Rips barcode lies within bottleneck distance δ_s of a “star diagram” read off the distances to s . Every stable vectorization is therefore a function of one column of the attention matrix, up to δ_s . Applied to TOHA, the same argument shows that its prompt–response topological divergence *equals* the mean, over response tokens, of one minus their largest prompt attention whenever a prompt-level coning defect vanishes, and lies within that defect otherwise. On 12,336,672 per-head graphs from 11 settings (5 open models of 1.1B–3.8B parameters; TruthfulQA, HaluEval, and an on-policy TriviaQA benchmark we build), this identity has no violations and is exact on 57–88% of them. Adding TOHA’s score to its first-order counterpart changes AUC by at most 0.007 (equivalent within ± 0.015 in 11 of 11 settings), and supervised probes on the two are equivalent in 11 of 11. For whole-graph PH, 568,028 layer graphs show no bound violations; in every model with a first-token sink, 36–88% are *exactly* coned, 1D persistence is zero on all of them, and length-normalized 0D persistence keeps a median per-layer rank correlation of at least 0.98 with sink attention. For detection, on TruthfulQA and on-policy TriviaQA, none of 0D PH, sink-deflated PH, or the per-head coning defect adds more than 0.008 AUC to cheap non-topological probes, and TOHA adds at most 0.013. We release the extractor, evaluator, and all per-example features.

1 INTRODUCTION

A growing family of hallucination detectors treats a transformer’s attention matrix as a weighted graph and summarizes it with persistent homology (PH): barcodes of connected components (H_0) and loops (H_1) along a Vietoris–Rips filtration (Kushnareva et al., 2021; Cherniavskii et al., 2022; Perez & Reinauer, 2022; Kostenok et al., 2023; Bazarova et al., 2026; Samaga et al., 2026). The appeal is that topology might capture *global* structure in how information flows that token-level statistics miss. A separate line of work has established that decoder-only language models route a large share of attention to the first token, the *attention sink* (Xiao et al., 2024; Sun et al., 2024; Cancedda, 2024; Gu et al., 2025; Barbero et al., 2025). Only recently have the two been connected: Binkowski et al. (2026) show that sinks carry much of the signal used by attention-based hallucination detectors. No work has yet asked what sinks do to the *topology* of attention graphs.

We show that, for topological detectors, the two literatures study the same object to a large and quantifiable extent. When one vertex s is closer to every other vertex than those vertices are to each other, every sublevel set of the Vietoris–Rips filtration is a cone with apex s : all higher homology vanishes and the minimum spanning tree (which computes H_0) is the star around s . Real attention graphs sit at or very near this configuration, because of the sink. The “topology” of a causal attention graph is therefore, in most layers, exactly a function of how much attention each token sends to the sink, and in the remaining layers is controlled by a single computable number, the coning defect.

The same holds for the published detector that restricts topology to the prompt–response interface: TOHA’s divergence is a first-order attention statistic plus a computable correction, and on real heads the correction adds at most 0.007 AUC.

Contributions.

1. **A whole-barcode sink reduction** (§3). The ingredients are classical: a flag complex with a dominating vertex is a cone (Barmak & Minian, 2012; Boissonnat & Pritam, 2020), and persistence is stable (Cohen-Steiner et al., 2007; Chazal et al., 2009; 2014). We combine them into one-sided bounds in a single computable number. For any symmetric dissimilarity and vertex s , bars of dimension $k \geq 1$ have length at most δ_s , and 0D total persistence is sandwiched around the star weight (Theorem 1). The whole barcode is within bottleneck distance δ_s of the star diagram, so every stable vectorization reduces to one column of distances (Corollary 1). For causal attention, δ_0 has a closed form (Corollary 2).
2. **The published topological detector is a first-order detector** (§3, §5.4). TOHA’s topological divergence equals one minus the response tokens’ mean max-prompt attention on prompt-coned heads, and is within a computable defect of it otherwise (Proposition 1). This makes exact the qualitative remark of Binkowski et al. (2026). On 12,336,672 real head graphs the identity has no violations, and adding TOHA to its first-order counterpart changes AUC by 0.000 to +0.007.
3. **Verification of the whole-graph reduction** (§5.1, Appendix D). There are zero bound violations in 568,028 graphs. Graphs are exactly coned in 36–88% of cases in sink-dominated models, H_1 is empty on every coned graph, and a sink dose-response experiment traces the weakness of 1D features to coning.
4. **A detection audit with strong baselines** (§5.3). On TruthfulQA, HaluEval, and an on-policy TriviaQA benchmark, we compare 0D/1D PH, sink features, sink-deflated PH, the coning defect, and TOHA against length, lexical, log-probability, LLM-Check (Sriramanan et al., 2024), Lookback-Lens (Chuang et al., 2024), and hidden-state probes, with grouped cross-validation, early- and late-fusion incremental tests, and equivalence tests.

2 BACKGROUND AND RELATED WORK

Topology of attention. Kushnareva et al. (2021) and Cherniavskii et al. (2022) introduced barcodes of symmetrized BERT attention graphs as features for artificial-text detection and acceptability; Perez & Reinauer (2022) and Kostenok et al. (2023) extended them to classification and uncertainty estimation. For hallucination detection, TOHA (Bazarova et al., 2026) computes an MST-based divergence between prompt and response subgraphs, HalluZig (Samaga et al., 2026) uses zigzag persistence across layers, and CHARM (Frasca et al., 2026) learns a message-passing model over attention graphs. Gardinazzi et al. (2024) study zigzag persistence of hidden representations. All 0D summaries in this family are MST statistics (Fact 1; Gower & Ross, 1969); our results concern what those statistics measure in *causal* models.

Attention sinks. Xiao et al. (2024) observed that streaming LMs keep the first tokens’ key-values because a large share of attention lands there; Sun et al. (2024) tied sinks to massive activations, Cancedda (2024) to low-rank “dark signals”, Gu et al. (2025) characterized when they emerge during pre-training, and Barbero et al. (2025) argued they prevent over-mixing; related “no-op” attention to special tokens was observed in BERT (Clark et al., 2019) and vision transformers (Darcet et al., 2024). Binkowski et al. (2026) use sinks with large-norm values directly as a hallucination signal (SinkProbe) and argue that earlier attention-based detectors rely on them implicitly. Our theorem makes this precise for topological detectors, turning sink strength into a statement about the topology of attention graphs.

Hallucination detection. Beyond topology, detectors use sampling consistency (Manakul et al., 2023; Farquhar et al., 2024), internal probes (Azaria & Mitchell, 2023; Burns et al., 2022; Marks & Tegmark, 2023; Li et al., 2023b; Kossen et al., 2024; Orgad et al., 2024; Du et al., 2024), covariance spectra (Chen et al., 2024), attention-kernel log-determinants (Sriramanan et al., 2024), and context-vs-generated attention ratios (Chuang et al., 2024). Recent audits find that several benchmarks admit near-perfect detection from surface form alone (Hussain & Kantarcioglu, 2026; Janiak et al., 2025),

echoing annotation artifacts in NLI (Gururangan et al., 2018; McCoy et al., 2019); we therefore include lexical controls and an on-policy benchmark.

3 THEORY: ATTENTION SINKS CONE THE VIETORIS–RIPS FILTRATION

Let $D \in \mathbb{R}_{\geq 0}^{N \times N}$ be symmetric with zero diagonal. The Vietoris–Rips filtration is $K_\epsilon = \{\sigma \subseteq [N] : \max_{u,v \in \sigma} D_{uv} \leq \epsilon\}$; $\text{dgm}_k(D)$ is the barcode of $H_k(K_\epsilon; \mathbb{F})$ over a field, and $P_k(D)$ is its total persistence (sum of finite bar lengths). For attention, $A^{(l)}$ is the head-averaged, row-stochastic, lower-triangular attention matrix of layer l and, following the literature, $D = 1 - \max(A, A^\top)$ with zero diagonal.

Fact 1 (Gower & Ross, 1969). *All 0D bars are born at 0, and their finite death times are, as a multiset, the edge weights of a minimum spanning tree (MST) of D . In particular $P_0(D) = w(\text{MST}(D))$.*

Definition 1 (Coning defect, star weight). For $s \in [N]$, the *coning defect* is $\delta_s(D) = \max_{u \neq v, u, v \neq s} [\max(D_{su}, D_{sv}) - D_{uv}]_+$ (zero if $N < 3$), and the *star weight* is $S_s(D) = \sum_{u \neq s} D_{su}$. We call D *coned at s* if $\delta_s(D) = 0$, i.e. s is at least as close to both endpoints of every edge as they are to each other.

Theorem 1 (Sink-coning reduction). *For every symmetric D and every vertex s , with $\delta = \delta_s(D)$:*

- (a) *every bar of $\text{dgm}_k(D)$, $k \geq 1$, has length at most δ ; hence $P_k(D) \leq \delta \cdot |\text{dgm}_k(D)|$;*
- (b) *$S_s(D) - (N-1)\delta \leq P_0(D) \leq S_s(D)$, and $\max_u D_{su} - \delta \leq \max \text{dgm}_0^{\text{fin}}(D) \leq \max_u D_{su}$;*
- (c) *if D is coned at s , then $H_k(K_\epsilon) = 0$ for all $k \geq 1$ and all ϵ , the star at s is an MST, and $P_0(D) = S_s(D)$.*

Proof sketch (full proof in Appendix A). Let $D'_{uv} = \max(D_{uv}, D_{su}, D_{sv})$ for $u, v \neq s$ and $D'_{su} = D_{su}$. Then $D \leq D' \leq D + \delta$ entrywise and D' is coned at s . Since D' is coned, any simplex $\sigma \not\ni s$ with $|\sigma| \geq 2$ has the same filtration value as $\sigma \cup \{s\}$, and a vertex not yet joined to s has no edges; so each K'_ϵ is a cone with apex s plus isolated points, and $H_k(K'_\epsilon) = 0$ for $k \geq 1$. The inclusions $K_b \subseteq K'_{b+\delta} \subseteq K_{b+\delta}$ factor the persistence map $H_k(K_b) \rightarrow H_k(K_{b+\delta})$ through zero, so no bar outlives δ . For (b), the star is a spanning tree of D (upper bound) and an MST of D' ; MST weight and bottleneck are $(N-1)$ - and 1-Lipschitz in the sup-norm. The ingredients are standard: Lemma 1 is the dominating-vertex (cone) argument behind strong and edge collapses (Barmak & Minian, 2012; Boissonnat & Pritam, 2020), and the last step is a one-sided form of the interleaving argument behind persistence stability (Chazal et al., 2009). $\square$

The bound in (a) is attained: if $D_{su} = 1$ for all u and four further vertices form a 4-cycle with side $\frac{1}{2}$ and diagonals 1, then $\delta_s = \frac{1}{2}$ and H_1 has the single bar $[\frac{1}{2}, 1)$.

Theorem 1 controls summaries; the same comparison controls the whole barcode. Call $\Sigma_s(D)$ the *star diagram*: in degree 0 the bars $[0, D_{su})$, $u \neq s$, plus one infinite bar, and nothing in degree $k \geq 1$.

Corollary 1 (The whole barcode is the star, up to δ_s). *For every k , $d_B(\text{dgm}_k(D), \Sigma_s(D)_k) \leq \delta_s$, with equality of the diagrams when $\delta_s = 0$. More precisely, the j -th smallest finite 0D death of D lies in $[D_{s(j)} - \delta_s, D_{s(j)}]$, where $D_{s(j)}$ is the j -th smallest entry of $(D_{su})_{u \neq s}$, and β_0 of K_ϵ lies between $1 + \#\{u : D_{su} > \epsilon + \delta_s\}$ and $1 + \#\{u : D_{su} > \epsilon\}$. Hence every vectorization that is L -Lipschitz in d_B (e.g. persistence landscapes; Bubenik, 2015) is within $L\delta_s$ of a function of the column $(D_{su})_u$. Vectorizations that are stable in 1-Wasserstein distance (e.g. persistence images; Adams et al., 2017) are within $L((N-1)\delta_s + \frac{1}{2}\delta_s \sum_{k \geq 1} |\text{dgm}_k(D)|)$ of such a function. On coned graphs, every vectorization, including persistence entropy and Betti curves, is exactly such a function.*

Corollary 2 (Causal attention). *Let A be lower-triangular and row-stochastic, $D = 1 - \max(A, A^\top)$, and $s = 0$ the first token. Then $D_{0u} = 1 - A_{u0}$, so $S_0 = (N-1)(1 - \bar{m})$ with $\bar{m} = \frac{1}{N-1} \sum_{u \geq 1} A_{u0}$ the mean sink attention, and*

$$\delta_0 = \max_{1 \leq v < u} [A_{uv} - \min(A_{u0}, A_{v0})]_+.$$

Thus $P_0 \in [(N-1)(1-\bar{m}-\delta_0), (N-1)(1-\bar{m})]$ and all $H_{k \geq 1}$ bars have length at most δ_0 . The graph is exactly coned iff, for all $1 \leq v < u$, $A_{uv} \leq A_{u0}$ and $A_{uv} \leq A_{v0}$: no token attends to an earlier non-sink token more than either of the two attends to the sink.

Remark 1 (Choice of distance). Coning is an order condition, so it is invariant under any entrywise strictly increasing transform f of D with $f(0) = 0$ (e.g. $-\log$ of the symmetrized attention, or ranks without ties, in place of $1 - A$). The set of exactly coned graphs, and the vanishing of their higher homology, does not depend on how attention is turned into a distance; only the magnitude of δ_s on non-coned graphs does. (A merely non-decreasing f , such as clipping, can create coning.)

Corollary 2 makes four testable predictions: **(C1)** in coned layers 0D total persistence equals $(N-1)(1-\bar{m})$, so after dividing by $N-1$ it is an affine function of sink attention; **(C2)** H_1 is empty whenever $\delta_0 = 0$; **(C3)** 1D persistence can only be non-zero where the sink is violated, with bars no longer than δ_0 ; **(C4)** removing the sink and renormalizing (*deflation*) exposes whatever non-sink topology exists.

Random weights and directed complexes. For i.i.d. weights, by contrast to Theorem 1, 1D total persistence grows linearly in N (Hiraoka & Shirai, 2017; Hino & Kanazawa, 2019); we give an elementary explicit bound, $\mathbb{E}[P_1] \geq \frac{3}{8}N - \sqrt{3N} + O(1)$ (Proposition 2, Appendix A). For causal attention the directed flag complex is no escape: the digraph is a transitive tournament, so its weighted directed flag complex is isomorphic to $\text{VR}(D)$ as a filtered complex (Proposition 3, Appendix A.4).

The published topological detector reduces to a first-order statistic. TOHA (Bazarova et al., 2026) scores an attention head by the *topological divergence* between the response tokens R and the prompt tokens P . Its per-head score is $d(R, P) = \text{MTopDiv}(R, P)/|R|$, where MTopDiv is the total 0D persistence of the head’s graph $D = 1 - \max(A, A^\top)$ after all prompt–prompt distances are set to 0 (equivalently, the minimum spanning forest attaching R to P). TOHA averages d over at most ten heads chosen on labelled data. For $u \in R$ let $\pi_u = \max_{v \in P} A_{uv}$ be u ’s largest attention to any prompt token.

Proposition 1 (TOHA is max-prompt attention, up to a defect). *Let A be lower-triangular and row-stochastic, and let $P = \{0, \dots, p-1\}$ and $R = \{p, \dots, N-1\}$. With $\delta_P = \max_{w < u \in R} [A_{uw} - \min(\pi_u, \pi_w)]_+$,*

$$\frac{1}{|R|} \sum_{u \in R} (1 - \pi_u) - \delta_P \leq d(R, P) \leq \frac{1}{|R|} \sum_{u \in R} (1 - \pi_u),$$

with equality when $\delta_P = 0$. Moreover $1 - \pi_u \leq 1 - A_{u0}$, with equality iff the first token is among u ’s most-attended prompt tokens. On a head where this holds for every $u \in R$ and $\delta_P = 0$, $d(R, P) = 1 - \frac{1}{|R|} \sum_{u \in R} A_{u0}$, the response tokens’ mean sink attention.

Proof sketch. Zeroing the prompt block merges P at scale 0, so the remaining finite 0D bars are the MST weights of the quotient graph on $\{p^*\} \cup R$, where $D_{p^*u} = \min_{v \in P} D_{uv} = 1 - \pi_u$ (Fact 1). Apply Theorem 1(b) at the apex p^* ; as in Corollary 2, its coning defect is δ_P and its star weight is $\sum_u (1 - \pi_u)$. Divide by $|R|$. $\square$

Binkowski et al. (2026) note that sink nodes are “likely to appear in the MST” of MTopDiv . Proposition 1 makes this exact, and it shows where the topology can enter: only through δ_P , which is non-zero only when a response token attends to an earlier response token more than one of the two attends to its best prompt token. We test the proposition on real heads, and TOHA against its first-order counterparts, in §5.4.

4 EXPERIMENTAL SETUP

Models and benchmarks. We use Qwen2.5-1.5B/3B-Instruct, Phi-3-mini-4k-instruct, TinyLlama-1.1B-Chat, and SmoLLM-1.7B-Instruct (Mistral-7B-Instruct results appear in Appendix G when available). **TruthfulQA** (Lin et al., 2022): all 817 questions, each paired with its best answer and first incorrect answer (1,634 rows) under each model’s chat format (Appendix B).

Table 1: **Theorem 1 on real attention.** *cells*: (example, layer) graphs; *viol.*: violations of any bound in Theorem 1(a)–(b) with $s = 0$; *coned*: share with $\delta_0 = 0$ exactly; $H_1 = \emptyset$ | *coned*: share of coned graphs with empty H_1 (C2 predicts 100%). Last two columns: median over layers of the Spearman correlation across examples between P_0 and S_0 (ρ_{raw}), and between $P_0/(N-1)$ and $1 - \bar{m} = S_0/(N-1)$ ($\rho_{/N}$), which removes the shared dependence on sequence length.

Benchmark	Model	$\bar{N}$	graphs	viol.	sink attn.	coned	$H_1 = \emptyset$ coned	ρ_{raw}	$\rho_{/N}$
TruthfulQA	Qwen2.5-3B	44.2	58,824	0	0.50	62.2%	100.0%	1.000	1.000
TruthfulQA	Qwen2.5-1.5B	44.2	45,752	0	0.53	64.5%	100.0%	1.000	1.000
TruthfulQA	Phi-3-mini	29.0	52,288	0	0.69	88.4%	100.0%	1.000	1.000
TruthfulQA	TinyLlama-1.1B	42.3	35,948	0	0.53	36.2%	100.0%	1.000	0.982
TruthfulQA	SmolLM-1.7B	32.4	39,216	0	0.02	0.0%	–	0.967	0.412
HaluEval	Qwen2.5-3B	126.9	72,000	0	0.42	52.7%	100.0%	1.000	1.000
HaluEval	Qwen2.5-1.5B	128.1	28,000	0	0.45	62.5%	100.0%	1.000	1.000
TriviaQA (on-policy)	Qwen2.5-3B	68.6	72,000	0	0.48	63.0%	100.0%	1.000	1.000
TriviaQA (on-policy)	Qwen2.5-1.5B	68.8	56,000	0	0.52	50.6%	100.0%	1.000	1.000
TriviaQA (on-policy)	Phi-3-mini	46.1	64,000	0	0.69	87.1%	100.0%	1.000	1.000
TriviaQA (on-policy)	TinyLlama-1.1B	72.0	44,000	0	0.50	36.4%	100.0%	1.000	0.991

HaluEval QA (Li et al., 2023a): the first 1,000 (Qwen2.5-3B) or 500 (Qwen2.5-1.5B) knowledge-grounded questions with right and hallucinated answers. Both are *teacher-forced*: the model reads dataset-written answers. **TriviaQA on-policy**: 2,000 random validation questions (Joshi et al., 2017); each model answers greedily (short-phrase instruction, ≤ 24 tokens), an answer is labelled hallucinated unless a normalized gold alias occurs in it, and features are extracted by re-reading the model’s *own* answer. Per-model accuracies are in Table 6. All models run in bfloat16 with eager attention.

Features. Per layer (head-averaged A): 0D (max lifetime, total persistence); 1D (four summaries, divided by N); SINK (S_0 and $\max_u D_{0u}$, i.e. exactly the two quantities Corollary 2 says 0D reduces to); DEFLATED PH (0D+1D after deleting token 0 and renormalizing rows); ANSWER-ONLY PH (induced subgraph on answer tokens); row entropy and row max. Per head: 0D (MST total and max via a dense Prim’s algorithm), sink mass, entropy, Lookback ratio (Chuang et al., 2024), and the LLM-Check attention score (Sriramanan et al., 2024). TOHA (per head, separate forward pass, same rows): the exact score d of Bazarova et al. (2026) via a dense Prim’s algorithm on the prompt-contracted graph, the $\bar{\pi}$ -score, the response sink score, and δ_P (Proposition 1). Other: answer-token log-probability and predictive-entropy statistics; mean-pooled answer hidden states at the middle layer; TF-IDF word and character n -grams of the answer text (LEXICAL).

Protocol. Every number uses one evaluator: standardized features and L_2 logistic regression, with $C=1$ for banks of at most 64 dimensions and otherwise $C \in \{0.01, 0.1, 1\}$ chosen by inner grouped 3-fold CV. The outer loop is 10-fold StratifiedGroupKFold grouped by question, repeated over 3 seeds (42–44), with out-of-fold probabilities averaged. Incremental value is tested in two ways. *Early fusion* (“X|Y”) concatenates the banks. *Late fusion* (“X $\oplus$ Y”) stacks the two banks’ out-of-fold logits in a second-level logistic regression under the same CV, so that a small bank cannot be drowned out by a wide one. Differences in pooled AUC use a 10,000-resample question-level cluster bootstrap. We call two banks *equivalent* if the 90% interval lies inside ± 0.015 AUC (two one-sided tests, TOST, at $\alpha=0.05$), and we report TOST power. Paired benchmarks also report within-pair accuracy. The theory checks use every (example, layer) graph. Ripser (Bauer, 2021) stores filtration values in 32-bit floats, so bar-length and max-death bounds are checked at tolerance 10^{-5} , and the P_0 sandwich, a sum of up to N such values, at 10^{-3} .

5 RESULTS

5.1 THE REDUCTION HOLDS ON REAL ATTENTION

Table 1 checks Theorem 1 on every layer-level graph of every example: 568,028 graphs in total. There are 0 violations of any bound. In the 10 settings whose models place 0.42 to 0.69 of attention on token 0, 36–88% of graphs are *exactly* coned ($\delta_0 = 0$), and on every one of them H_1 is empty, as prediction C2 requires. Across graphs, δ_0 and 1D total persistence have Spearman correlation 0.32 to 0.89 (C3). Both P_0 and the star weight $S_0 = (N-1)(1 - \bar{m})$ scale with sequence length, so their raw per-layer rank correlation (median ≥ 0.999) partly reflects length. After dividing both by $N-1$,

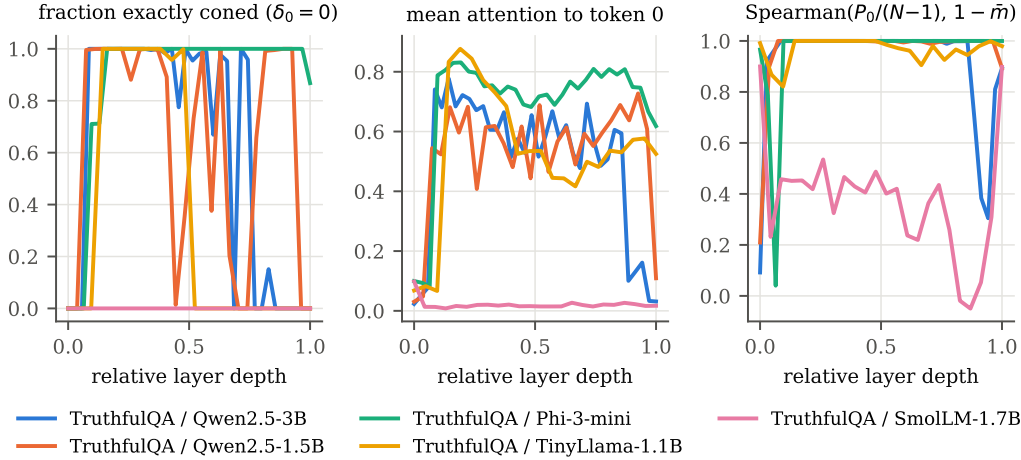


Figure 1: **Layerwise structure (TruthfulQA).** Coning (left) tracks sink attention (middle): layers with little sink attention, typically the first few and, in some models, the last few, are rarely coned, and SmolLM-1.7B, which has no first-token sink, is never coned. Right: length-normalized rank correlation between P_0 and sink attention. It is near 1 in the layers of sink models, dips in some open layers, and is low throughout SmolLM-1.7B.

the median per-layer rank correlation between $P_0/(N-1)$ and $1 - \bar{m}$ is still at least 0.98 in every sink setting (C1). In most layers, then, length-normalized 0D total persistence is, up to rank, one minus the mean attention paid to the first token. The weakest single layer per setting reaches only -0.14 to 0.84.

Figure 1 shows where coning happens: in the middle of the network, where sink attention is high. The first few layers are never sink-dominated, and in some models (TinyLlama, the Qwen models) so is a block of late layers; these layers are rarely coned. There, the length-normalized correlation can drop, as Theorem 1(b) allows once $(N-1)\delta_0$ is comparable to the spread of S_0 across examples.

A model without a first-token sink as a contrast. SmolLM-1.7B, under its ChatML format, places only 0.02 of its attention on token 0, and no vertex cones any of its graphs (the best apex never reaches $\delta_s = 0$). Theorem 1 then gives no useful guarantee. Its H_1 is non-empty on 87% of graphs, and although the raw $\rho(P_0, S_0)$ is 0.967, this is almost entirely shared length: the length-normalized correlation is 0.41. SmolLM differs from the other models in weights, data and template, so it is a contrast, not a controlled intervention; the within-model intervention is deflation (C4, below). The reduction is a property of sink-dominated attention, not of the pipeline.

5.2 WHAT TOPOLOGY ADDS BEYOND THE SINK

If 0D persistence reduced *exactly* to the star, a classifier on the two sink features would match one on the two 0D features in every layer. Table 3 (“Sink vs. 0D”) shows the two banks differ by -0.015 to +0.031 AUC across the 10 sink settings evaluated, and are statistically equivalent within ± 0.015 in 5 of them (the TOST power of this comparison is modest, so non-equivalence is not evidence of a difference). Adding 0D to the sink bank gains +0.003 to +0.028.

The residual lives exactly where the theory says it can. We split layers without using labels: *coned* layers ($\geq 90\%$ of graphs coned) and *open* layers ($< 50\%$). Adding the coned layers’ 0D features to SINK changes AUC by -0.001 to +0.004, whereas the open layers account for +0.003 to +0.027 (Appendix E). Almost everything 0D adds beyond sink attention comes from the layers where the sink does not dominate.

Per head, the reduction is tighter. Per-head sink mass and per-head 0D are far stronger than their head-averaged versions (Table 2), but adding per-head MST total ($P_{0,h}$) to per-head sink mass

Table 2: **Pooled out-of-fold ROC-AUC** (3-seed repeated grouped 10-fold CV; seed SD ≤ 0.008). TQA: TruthfulQA; HE: HaluEval; TrQA: on-policy TriviaQA. Lex.: answer-text TF-IDF; LogP: log-probability statistics; LLM-C.: LLM-Check; Sink: $(S_0, \max_u D_{0u})$ per layer; 0D/1D: persistence per layer; subscript h : per head (Sink $_h$: sink mass; 0D $_h$: MST total and max); Lookb.: Lookback ratios; Hidden: mid-layer probe; Non-topo.: LogP+LLM-C.+Lookb.+Hidden+row statistics. Best per row in bold. Full table in Appendix E.

Setting	Lex.	LogP	LLM-C.	Sink	0D	1D	Sink $_h$	0D $_h$	Lookb.	Hidden	Non-topo.
TQA Qw3B	.820	.630	.693	.737	.736	.660	.851	.853	.897	.913	.926
TQA Qw1.5B	.820	.638	.666	.723	.726	.586	.830	.838	.851	.916	.917
TQA Phi-3	.820	.578	.653	.651	.646	.526	.871	.868	.892	.933	.936
TQA TinyLL	.820	.637	.614	.643	.654	.592	.808	.806	.850	.883	.885
TQA SmolLM	.820	.588	.604	.611	.583	.543	.720	.800	.863	.894	.899
HE Qw3B	.981	.993	.745	.829	.811	.710	.958	.971	.998	1.00	1.00
HE Qw1.5B	.979	.995	.791	.820	.789	.747	.922	.949	.999	1.00	1.00
TrQA Qw3B	.596	.839	.688	.717	.732	.601	.820	.807	.861	.841	.870
TrQA Qw1.5B	.609	.823	.676	.707	.716	.574	.767	.768	.824	.855	.873
TrQA Phi-3	.635	.877	.676	.707	.710	.546	.826	.807	.840	.867	.883
TrQA TinyLL	.740	.724	.688	.720	.720	.659	.814	.775	.823	.827	.842

changes AUC by only -0.001 to +0.011, which is equivalent within ± 0.015 in 10 of the 10 sink settings with per-head features.

The coning defect is a signal, but not a new one. Theorem 1 singles out δ_0 as the only place where topology can depart from the sink, so we also test it as a detector. Head-averaged, it is weak (AUC 0.50 to 0.65). Per head it is strong (0.69 to 0.96), and stacked onto per-head sink mass it adds +0.006 to +0.027 AUC (95% interval above 0 in 11 of 11 settings). Stacked onto the non-topological probes, however, it adds only 0.000 to +0.008 (above 0 in 3, equivalent within ± 0.015 in 11 of 11).

Deleting the sink exposes some, redundant, structure. Deflated PH (C4) adds +0.029 to +0.045 over SINK on TruthfulQA (95% interval excludes 0 in 4 of 4 sink settings) but only +0.002 to +0.029 on the on-policy benchmark (1 of 4). Answer-only PH behaves similarly (Appendix E). So attention graphs carry some non-sink geometry, most clearly on teacher-forced text. As §5.3 shows, this geometry is already available to non-topological probes.

One-dimensional persistence. Most real graphs have no 1D bars at all (Table 1), unlike i.i.d. weights, where 1D persistence grows linearly. Adding 1D to 0D changes AUC by -0.008 to +0.033 and is equivalent within ± 0.015 in 7 of 10 sink settings. A sink dose-response experiment and a planted-cycle experiment (Appendix D) show why: under a strong sink, a loop is visible to H_1 only if it breaks the cone.

5.3 HALLUCINATION DETECTION AGAINST STRONG BASELINES

Table 2 places attention-graph topology among standard detectors. Three patterns hold across benchmarks.

Topology is dominated by cheap non-topological probes. On TruthfulQA and the on-policy benchmark, a mean-pooled hidden-state probe (0.83 to 0.93) and per-head Lookback ratios (0.82 to 0.90) outperform every topological bank, and their combination with log-probability, LLM-Check, and row statistics reaches 0.84 to 0.94. Adding 0D PH to that combination changes AUC by 0.000 to +0.001 (equivalent within ± 0.015 in 11 of the 11 settings with per-head features), and adding sink-deflated PH changes it by -0.001 to +0.002 (equivalent in 11 of 11). Early fusion could hide a small bank inside a $\sim 2,000$ -dimensional one, so we repeat the test with late fusion. Stacking the 0D score onto the non-topological score changes AUC by 0.000 to +0.002 (equivalent in 11 of 11 settings; the 95% interval lies above 0 in 0 of them), and stacking the deflated-PH score changes it by 0.000 to +0.001 (equivalent in 11 of 11).

Which cheap probe wins depends on whether text is teacher-forced. On TruthfulQA, token log-probability is weak (0.58 to 0.64), consistent with the benchmark’s design around imitative false-

Table 3: **Key paired comparisons** (Δ AUC). “X–Y” is $\text{AUC}(X) - \text{AUC}(Y)$; “X|Y” is $\text{AUC}(Y+X) - \text{AUC}(Y)$ under early fusion; “X $\oplus$ Y” is the same under late fusion (stacked out-of-fold logits). $P_{0,h}$: per-head MST total. Defl.: sink-deflated 0D+1D PH; NT: all non-topological features. $\equiv$: equivalent within ± 0.015 (90% interval inside the margin); *: 95% interval excludes 0. Intervals and TOST power for every cell are in Appendix E.

Setting	Sink–0D	0D Sink	$P_{0,h}$ Sink _h	Defl. Sink	1D 0D	0D NT	0D $\oplus$ NT	Defl. NT
TQA Qw3B	+0.001 $\equiv$	+0.003 $\equiv$	+0.009 $\equiv$	+0.029*	+0.033*	0.000 $\equiv$	0.000 $\equiv$	+0.001 $\equiv$
TQA Qw1.5B	–0.003 $\equiv$	+0.012*	+0.011 $\equiv$	+0.034*	–0.008 $\equiv$	0.000 $\equiv$	0.000 $\equiv$	0.000 $\equiv$
TQA Phi-3	+0.005 $\equiv$	+0.006 $\equiv$	0.000 $\equiv$	+0.033*	–0.003 $\equiv$	0.000 $\equiv$	0.000 $\equiv$	+0.001 $\equiv$
TQA TinyLL	–0.010	+0.020*	+0.003 $\equiv$	+0.045*	+0.007 $\equiv$	0.000 $\equiv$	0.000 $\equiv$	–0.001 $\equiv$
TQA SmoLLM	+0.028*	+0.030*	+0.106*	+0.041*	+0.005 $\equiv$	0.000 $\equiv$	0.000 $\equiv$	–0.001 $\equiv$
HE Qw3B	+0.018*	+0.028*	+0.004 $\equiv$	+0.058*	+0.015*	0.000 $\equiv$	0.000 $\equiv$	0.000 $\equiv$
HE Qw1.5B	+0.031*	+0.012*	+0.008 $\equiv$	+0.052*	+0.026*	0.000 $\equiv$	0.000 $\equiv$	0.000 $\equiv$
TrQA Qw3B	–0.015*	+0.021*	+0.004 $\equiv$	+0.002	–0.003 $\equiv$	+0.001 $\equiv$	+0.002 $\equiv$	0.000 $\equiv$
TrQA Qw1.5B	–0.009	+0.010	+0.004 $\equiv$	+0.012	–0.002 $\equiv$	0.000 $\equiv$	0.000 $\equiv$	–0.001 $\equiv$
TrQA Phi-3	–0.004 $\equiv$	+0.006 $\equiv$	+0.002 $\equiv$	+0.029*	–0.005 $\equiv$	0.000 $\equiv$	0.000 $\equiv$	+0.002 $\equiv$
TrQA TinyLL	+0.001 $\equiv$	+0.004 $\equiv$	–0.001 $\equiv$	+0.014	–0.002 $\equiv$	0.000 $\equiv$	0.000 $\equiv$	0.000 $\equiv$

Table 4: **TOHA against its first-order counterparts** (Proposition 1). $\delta_P=0$: share of (example, head) graphs that are exactly prompt-coned, on which d equals the $\bar{\pi}$ -score. $\rho(d, \bar{\pi})$: median over heads of the rank correlation across examples. AUC columns: TOHA’s head-selection algorithm run on d (TOHA), on the $\bar{\pi}$ -score (TOHA[$\bar{\pi}$]), and on the response sink score (TOHA[$\bar{a}_0$]); then supervised probes on all heads of each score; then the non-topological bank. Δ columns and marks as in Table 3; $\oplus$ is late fusion.

Setting	$\delta_P=0$	$\rho(d, \bar{\pi})$	TOHA	TOHA[$\bar{\pi}$]	TOHA[$\bar{a}_0$]	TOHA _{sup}	$\bar{\pi}_{\text{sup}}$	$\bar{a}_{0,\text{sup}}$	Non-topo.	$\bar{\pi}$ –TOHA	$\bar{a}_0$ –TOHA	TOHA $\oplus\bar{\pi}$	TOHA $\oplus$ NT
TQA Qw3B	57%	.976	.668	.670	.649	.892	.892	.892	.926	+0.003 $\equiv$	–0.018*	+0.003 $\equiv$	+0.003 $\equiv$
TQA Qw1.5B	58%	.978	.646	.649	.628	.858	.857	.857	.917	+0.003 $\equiv$	–0.018*	+0.005 $\equiv$	+0.002 $\equiv$
TQA Phi-3	74%	.996	.751	.742	.715	.898	.901	.905	.936	–0.009*	–0.036*	0.000 $\equiv$	0.000 $\equiv$
TQA TinyLL	74%	.995	.666	.655	.618	.845	.843	.848	.885	–0.012*	–0.049*	+0.002 $\equiv$	+0.001 $\equiv$
TQA SmoLLM	58%	.954	.701	.695	.595	.862	.860	.841	.899	–0.007 $\equiv$	–0.107*	+0.005 $\equiv$	+0.002 $\equiv$
HE Qw3B	68%	.987	.967	.982	.961	.999	.999	1.00	1.00	+0.015*	–0.007 $\equiv$	0.000 $\equiv$	0.000 $\equiv$
HE Qw1.5B	69%	.988	.970	.988	.972	.999	.999	.999	1.00	+0.018*	+0.002 $\equiv$	0.000 $\equiv$	0.000 $\equiv$
TrQA Qw3B	76%	.991	.728	.736	.742	.862	.863	.870	.870	+0.009	+0.014*	+0.002 $\equiv$	+0.012*
TrQA Qw1.5B	77%	.992	.712	.714	.708	.828	.827	.836	.873	+0.002 $\equiv$	–0.005	+0.003 $\equiv$	+0.004 $\equiv$
TrQA Phi-3	88%	.998	.724	.731	.733	.847	.851	.850	.883	+0.007 $\equiv$	+0.008	0.000 $\equiv$	+0.001 $\equiv$
TrQA TinyLL	66%	.994	.714	.725	.693	.833	.826	.832	.842	+0.011*	–0.021*	+0.007 $\equiv$	+0.010*

hoods, and 0D PH (0.58 to 0.74) scores higher for 4 of 5 models, significantly so for 3. On-policy, log-probability is strong (0.72 to 0.88) and significantly beats 0D (0.71 to 0.73) for 3 of 4 models. Conclusions about topology versus confidence drawn from a single teacher-forced benchmark do not transfer.

Per-head resolution matters more than topology. Per-head features of every kind are far stronger than head-averaged ones, and per-head entropy lands within -0.016 to +0.034 AUC of per-head 0D (Appendix E).

5.4 THE PUBLISHED TOPOLOGICAL DETECTOR IS FIRST-ORDER

We test Proposition 1 on every head of every example: 12,336,672 head graphs from 11 settings (Table 4). We then run TOHA’s head-selection algorithm (Bazarova et al., 2026, Alg. 1) inside our cross-validation, selecting heads on the whole training fold, which is larger than TOHA’s probe set. We run it once on TOHA’s score d and, unchanged, on the two first-order scores the proposition names: the $\bar{\pi}$ -score $\frac{1}{|R|} \sum_u (1 - \pi_u)$ and the response sink score $1 - \bar{a}_0 = \frac{1}{|R|} \sum_u (1 - A_{u0})$.

The identity holds on real heads. The sandwich has 0 violations. Per setting, 57–88% of head graphs are exactly prompt-coned, and on those d equals the $\bar{\pi}$ -score up to float rounding (largest gap $6 \cdot 10^{-8}$). Across examples, the median per-head rank correlation between d and the $\bar{\pi}$ -score is 0.954 to 0.998. With the sink score it is 0.78 to 0.99 in the sink models and near zero for SmoLLM, which has no first-token sink. Shifting the sink’s attention logit moves these quantities as predicted (Appendix F).

Topology adds almost nothing to the first-order score. Stacking d onto the $\bar{\pi}$ -score changes AUC by 0.000 to +0.007. The change is equivalent to zero within ± 0.015 in 11 of 11 settings, although the 95% interval lies above 0 in 4 of them. With supervised probes on all heads (the setting of TOHA’s own feature comparison), the $\bar{\pi}$ -score differs from d by -0.007 to +0.004 (equivalent in 11 of 11). The response sink score differs from d by -0.021 to +0.008 (equivalent in 9). The exceptions are SmolLM, which has no sink and where the sink score is worse, and one on-policy setting whose interval extends past the margin on the positive side. TOHA’s discrete head selection is noisier: run on the $\bar{\pi}$ -score it changes AUC by -0.012 to +0.018 (significantly higher in 3 settings, lower in 2, non-inferior at margin 0.015 in 9 of 11). Run on the sink score it loses up to 0.049 AUC in the sink models (0.107 in SmolLM): the selected heads are prompt-coned on 70–95% of examples, but the sink is every response token’s top prompt token on only 19–79%. Stacked onto the non-topological probes, TOHA changes AUC by 0.000 to +0.012 (95% interval above 0 in 3 of 11); by the above, whatever it adds is carried by the $\bar{\pi}$ -score.

Why “attention to $\langle s \rangle$ ” looked weak. Bazarova et al. (2026, Table 9) report that attention to the first token is a much weaker feature than their divergence. Proposition 1 identifies the quantity that d actually tracks: per head, the attention of the *response* tokens to their most-attended prompt token. In the sink models, that token is the sink for every response token in 40–75% of head graphs.

Benchmark artifacts. A TF-IDF model on the answer text alone reaches 0.82 AUC on TruthfulQA and 0.98 on HaluEval, but only 0.60 to 0.74 on-policy: teacher-forced benchmarks reward recognizing how the dataset’s answers were *written* (Hussain & Kantarcioglu, 2026; Janiak et al., 2025). We draw no conclusions from HaluEval, where lexical separability is near perfect.

6 DISCUSSION

What attention-graph topology measures. In decoder-only models, Vietoris–Rips persistence of an attention graph, and every stable summary of it, is controlled by one scalar per graph: the coning defect of the first token. The same is true of TOHA’s prompt–response divergence, with the prompt playing the role of the apex. Where it is zero, as in most middle-layer graphs of every sink-dominated model we tested, 0D total persistence equals $(N-1)(1 - \bar{m})$, a function of length and sink attention alone, and higher homology is empty. Where it is positive, Theorem 1 bounds how far topology can depart from the sink; in our data this happens mainly in the first layers.

Recommendations. (i) Report sink attention and δ_0 (or the relevant defect, such as δ_P for TOHA) next to any attention-graph topological feature, and compare the detector with the first-order statistic that the reduction names. (ii) Compare topological detectors against per-head non-topological statistics (sink mass, entropy, Lookback ratios) at the same resolution, and against a hidden-state probe. (iii) Evaluate on the model’s own generations; teacher-forced benchmarks mix detection with recognition of dataset authorship. (iv) If structure beyond the sink is the goal, remove the sink first (deflation) or restrict to answer tokens. Both expose real non-sink geometry, but in our settings it was not new information.

Limitations. Our models range from 1.1B to 3.8B parameters (7B in the appendix when available); sink strength changes with scale and training (Gu et al., 2025), so the *share* of coned layers is model-specific, although Theorem 1 is not. Our results cover Vietoris–Rips persistence of symmetrized attention (head-averaged or per head), its causal directed-flag equivalent, every stable vectorization of it, and TOHA. They do not cover HalluZig (Samaga et al., 2026), which computes zigzag persistence of the *graph* (1-skeleton) homology of top-percentile edge sets across layers: a cone graph still has cycles, because its triangles are not filled, so Theorem 1 does not constrain its H_1 . Learned graph models (Frasca et al., 2026) are also outside the theorem, although their inputs are the same sink-dominated graphs. TOHA was designed for RAG; only HaluEval supplies a context passage here. Our on-policy labels come from alias string matching, which mislabels some paraphrases and hedged answers and has not been audited against human judgments (Janiak et al., 2025). TinyLlama’s on-policy answers were generated without the BOS token added at feature extraction. Our intervals resample questions for a fixed, seed-averaged predictor and do not include variation from refitting, so equivalence claims are conditional on the trained models. Prompt formats differ across models (Appendix B), so cross-model comparisons mix architecture and template.

REPRODUCIBILITY STATEMENT

All code is released at <https://anonymous.4open.science/r/ICLR-BE07/sinktda/extract.py> computes all whole-graph and per-head features, `sinktda/toha.py` computes TOHA and its first-order counterparts and runs TOHA’s head selection, and `sinktda/evaluate.py` produces every AUC, interval and theory check. `sinktda/late_fusion.py` and `sinktda/defect_probe.py` run the stacked tests, `sinktda/check_theory.py` and `sinktda/synthetic.py` run the numerical and synthetic checks, and `sinktda/report.py` generates every table, figure and number in the text. The run order is in `sinktda/run_fix.sh` and `sinktda/run_toha.sh`. Per-example features for all settings and the TriviaQA generations are included. Seeds, prompts, dtype and model identifiers are listed in Appendix B; proofs are in Appendix A.

LLM USAGE STATEMENT

LLMs were used to help implement experiment code, check reported statistics against result files, and draft and edit text. The authors verified every claim, number, proof, and citation and take full responsibility for the paper.

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A PROOFS

Throughout, D is a symmetric $N \times N$ matrix with non-negative entries and zero diagonal, K_ϵ is the Vietoris–Rips complex at scale ϵ , $\text{filt}_D(\sigma) = \max_{u,v \in \sigma} D_{uv}$ (with $\text{filt}_D(\{u\}) = 0$), and homology is taken over a field. Bars are half-open intervals $[b, d)$.

A.1 FACT 1

All vertices enter at 0. Processing edges in non-decreasing weight order, an edge either joins two components, in which case (elder rule) one 0D class dies at its weight and Kruskal’s algorithm adds the edge to the spanning forest, or it closes a cycle, in which case no 0D class dies and Kruskal discards it. Hence the $N - 1$ finite death times are the MST edge weights (Gower & Ross, 1969; Edelsbrunner & Harer, 2010).

A.2 THEOREM 1

Lemma 1 (Coned filtrations). *If D is coned at s , then for every $\epsilon \geq 0$, $K_\epsilon = C_\epsilon \sqcup I_\epsilon$, where C_ϵ is a simplicial cone with apex s and I_ϵ is a set of isolated vertices. Consequently $\tilde{H}_k(K_\epsilon) = 0$ for every $k \geq 1$.*

Proof. Let $\sigma \in K_\epsilon$ with $s \notin \sigma$ and $|\sigma| \geq 2$. For each $u \in \sigma$ choose $v \in \sigma \setminus \{u\}$; coning gives $D_{su} \leq D_{uv} \leq \text{filt}_D(\sigma) \leq \epsilon$. Therefore $\text{filt}_D(\sigma \cup \{s\}) = \text{filt}_D(\sigma)$ and $\sigma \cup \{s\} \in K_\epsilon$. Now let $u \neq s$ be a vertex with $D_{su} > \epsilon$. If u had an edge $\{u, v\} \in K_\epsilon$ then $D_{su} \leq D_{uv} \leq \epsilon$, a contradiction; so u is isolated. Let I_ϵ be these vertices and C_ϵ the subcomplex on the remaining vertices. Every vertex of C_ϵ is adjacent to s , and by the first part every simplex of C_ϵ extends by s , so C_ϵ is the cone over its link of s and is contractible. H_k of a disjoint union is the direct sum, and isolated points have no homology in degree ≥ 1 . $\square$

Proof of Theorem 1. Fix s and $\delta = \delta_s(D)$. Define $D'_{uv} = \max(D_{uv}, D_{su}, D_{sv})$ for $u, v \neq s$, $u \neq v$; $D'_{su} = D'_{us} = D_{su}$; $D'_{uu} = 0$.

Step 1: $D \leq D' \leq D + \delta$ and D' is coned at s . The first inequality is immediate. For $u, v \neq s$, $D'_{uv} - D_{uv} = [\max(D_{su}, D_{sv}) - D_{uv}]_+ \leq \delta$. Coning: $\max(D'_{su}, D'_{sv}) = \max(D_{su}, D_{sv}) \leq D'_{uv}$.

Step 2: interleaving. Since filtration values are maxima of entries, $\text{filt}_D \leq \text{filt}_{D'} \leq \text{filt}_D + \delta$. Hence for every b , $K_b \subseteq K'_{b+\delta} \subseteq K_{b+\delta}$, where K' is the filtration of D' .

Step 3: (a). The persistence map $H_k(K_b) \rightarrow H_k(K_{b+\delta})$ is induced by the composite inclusion in Step 2 and therefore factors through $H_k(K'_{b+\delta})$, which is zero for $k \geq 1$ by Lemma 1. The rank of $H_k(K_b) \rightarrow H_k(K_{b'})$ equals the number of bars $[b_i, d_i)$ with $b_i \leq b$ and $d_i > b'$. A bar with $d_i - b_i > \delta$ would contribute to this rank at $b = b_i$, $b' = b_i + \delta$, a contradiction. So every bar has length at most δ , and summing gives the bound on P_k .

Step 4: (b). By Fact 1, $P_0(D) = w_D(T_D)$ for an MST T_D of D . The star $T_s = \{\{s, u\}\}_{u \neq s}$ is a spanning tree, so $P_0(D) \leq w_D(T_s) = S_s(D)$. For the lower bound, T_s is an MST of D' , even with ties. Root any spanning tree T at s and let e_u be the edge from $u \neq s$ to its parent $p(u)$. If $p(u) = s$ then $D'_{e_u} = D'_{su}$; otherwise $D'_{e_u} = D'_{up(u)} \geq \max(D'_{su}, D'_{sp(u)}) \geq D'_{su}$ because D' is coned. The map $u \mapsto e_u$ is a bijection onto the edges of T , so $w_{D'}(T) \geq \sum_{u \neq s} D'_{su} = w_{D'}(T_s)$ and $\max_{e \in T} D'_e \geq \max_u D'_{su}$. Hence $w_{D'}(T_{D'}) = w_{D'}(T_s) = S_s(D)$, and $P_0(D) = w_D(T_D) \geq w_{D'}(T_D) - (N-1)\delta \geq w_{D'}(T_{D'}) - (N-1)\delta = S_s(D) - (N-1)\delta$. The maximum finite 0D death is the bottleneck value $\min_T \max_{e \in T} w(e)$, which is attained by any MST; it is at most $\max_u D_{su}$ (the star is a spanning tree) and is 1-Lipschitz in the sup-norm, which gives the lower bound because the bottleneck of D' is $\max_u D_{su}$ (by the rooted-tree argument above).

Relation to known results. Lemma 1 is the classical observation that a flag complex in which one vertex is adjacent to every non-isolated vertex, and to every simplex, is a cone; this is the dominated-vertex mechanism behind strong collapses (Barmak & Minian, 2012) and edge collapses of flag filtrations (Boissonnat & Pritam, 2020). Step 3 is a one-sided instance of the interleaving argument

behind the stability of persistence (Cohen-Steiner et al., 2007; Chazal et al., 2009; 2014). The two-sided stability theorem would give bars of length at most 2δ , whereas the one-sided comparison $D \leq D'$ gives δ , which is attained (Section 3).

Step 5: (c). If $\delta = 0$ then $D' = D$, and the claims follow from Lemma 1 and Step 4. $\square$

Proof of Corollary 1. Let D' be as in the proof of Theorem 1. Since D' is coned at s , Lemma 1 and Step 4 give $\text{dgm}_k(D') = \Sigma_s(D)_k$ for all k : no bars in degree $k \geq 1$, and 0D deaths equal to the star weights $D'_{su} = D_{su}$. For $1 \leq j \leq N - 1$, the j -th smallest finite 0D death of a matrix E is $w_j(E) = \min\{\epsilon : \beta_0(K_\epsilon^E) \leq N - j\}$. If $E \leq E'$ entrywise then $K_\epsilon^{E'} \subseteq K_\epsilon^E$ on the same vertex set, so $\beta_0(K_\epsilon^E) \leq \beta_0(K_\epsilon^{E'})$ and $w_j(E) \leq w_j(E')$; and $w_j(E + c) = w_j(E) + c$ on off-diagonal entries. From $D \leq D' \leq D + \delta_s$ we get $w_j(D) \leq D_{s(j)} \leq w_j(D) + \delta_s$. The same inclusions give the Betti-0 sandwich, because $\beta_0(K'_\epsilon) = 1 + \#\{u : D_{su} > \epsilon\}$. Matching the j -th deaths shows that $d_B(\text{dgm}_0(D), \Sigma_s(D)_0) \leq \delta_s$ and $W_1 \leq (N - 1)\delta_s$. In degree $k \geq 1$, every bar has length at most δ_s (Theorem 1(a)), so matching it to the diagonal costs at most $\delta_s/2$ in ℓ^∞ . The Lipschitz statements follow from the stability of landscapes in d_B (Bubenik, 2015) and of persistence images in W_1 (Adams et al., 2017). If $\delta_s = 0$ then $D = D'$. $\square$

Proof of Proposition 1. Let $\tilde{D}$ equal D except that $\tilde{D}_{vv'} = 0$ for $v, v' \in P$. At scale 0, Kruskal's algorithm joins all of P into one component with $|P| - 1$ zero-length bars. Every later merge is a merge in the quotient graph Q on $\{p^*\} \cup R$ with $Q_{p^*u} = \min_{v \in P} \tilde{D}_{vu}$ and $Q_{uw} = D_{uw}$. So $\text{MTopDiv}(R, P) = P_0(Q)$. For $u \in R$ and $v \in P$ we have $v < u$ and $D_{uv} = 1 - A_{uv}$, so $Q_{p^*u} = 1 - \pi_u$. For $w < u$ in R , $Q_{uw} = 1 - A_{uw}$, and $\max(Q_{p^*u}, Q_{p^*w}) - Q_{uw} = A_{uw} - \min(\pi_u, \pi_w)$, so $\delta_{p^*}(Q) = \delta_P$. Theorem 1(b) with $N - 1 = |R|$ gives the sandwich, with equality when $\delta_P = 0$ by part (c). Finally $\pi_u \geq A_{u0}$ because $0 \in P$. $\square$

We checked both statements numerically. The fast MTop-Div (a dense Prim's algorithm on Q) agrees with ripser run on $\tilde{D}$ to $3 \cdot 10^{-8}$ on 400 random causal matrices. On these, and on 300 further random matrices checked for the elementwise death bound and the bar bound in degrees 1 and 2, there are 0 violations (sinktda/check_theory.py).

Proof of Corollary 2. For $u > v$, $A_{vu} = 0$, so $D_{uv} = 1 - A_{uv}$; in particular $D_{0u} = 1 - A_{u0}$. For $1 \leq v < u$, $\max(D_{0u}, D_{0v}) - D_{uv} = A_{uv} - \min(A_{u0}, A_{v0})$. Substitute into Theorem 1. $\square$

A.3 LINEAR GROWTH OF 1D PERSISTENCE UNDER I.I.D. WEIGHTS

Length normalization of 1D features is usually motivated by the growth of cycles with N . For i.i.d. weights this growth is known: 0D total persistence converges to $\zeta(3)$ (Frieze, 1985), and 1D total persistence of the random clique-complex process is $\Theta(N)$ (Hiraoka & Shirai, 2017; Hino & Kanazawa, 2019). The following elementary bound makes the constant explicit.

Proposition 2 (Linear growth under i.i.d. weights). *If $\{D_{uv}\}_{u < v}$ are i.i.d. Uniform $[0, 1]$, then $\mathbb{E}[P_1(N)] \geq \int_0^1 \left[\binom{N}{2}p - N - \binom{N}{3}p^3 \right]_+ dp = \frac{3}{8}N - \sqrt{3N} + O(1)$. (A matching $O(N)$ upper bound is Theorem 1.5 of Hino & Kanazawa, 2019.)*

Proof. With i.i.d. uniform weights, the edges with $D_{uv} \leq p$ form an Erdős-Rényi graph $G(N, p)$ and K_p is its clique (flag) complex. Every 1D bar is contained in $[0, 1]$, because K_1 is the full simplex. Writing $\beta_1(p)$ for the first Betti number of K_p , the number of bars alive at p is $\beta_1(p)$, so $P_1 = \int_0^1 \beta_1(p) dp$. For any finite complex with f_i faces of dimension i , the strong Morse inequality at degree 2 reads $\beta_2 - \beta_1 + \beta_0 \leq f_2 - f_1 + f_0$ (Edelsbrunner & Harer, 2010), hence $\beta_1 \geq f_1 - f_0 - f_2 + \beta_0 + \beta_2 \geq f_1 - f_0 - f_2$. Since also $\beta_1 \geq 0$, Jensen's inequality gives $\mathbb{E}\beta_1(p) \geq [\mathbb{E}f_1 - f_0 - \mathbb{E}f_2]_+ = \left[\binom{N}{2}p - N - \binom{N}{3}p^3 \right]_+$. Integrating over p gives the first inequality. With $p = x/\sqrt{N}$ and $dp = dx/\sqrt{N}$, the integrand divided by $\sqrt{N}$ is $N(x/2 - x^3/6) - \sqrt{N} + O(1)$ uniformly for $x \in [0, \sqrt{3} + 1]$. It is positive on an interval whose endpoints are $O(N^{-1/2})$ from 0 and $\sqrt{3}$, and it is negative for larger x . Hence $\int_0^1 [\cdot]_+ dp = N \int_0^{\sqrt{3}} \left(\frac{x}{2} - \frac{x^3}{6} \right) dx - \sqrt{3N} + O(1) = \frac{3}{8}N - \sqrt{3N} + O(1)$. Numerically, $(\frac{3}{8}N - \text{bound})/\sqrt{N} = 1.62, 1.67, 1.70, 1.72$ at $N = 2^8, 2^{10}, 2^{12}, 2^{14}$, approaching

$\sqrt{3} \approx 1.73$. The order $\Theta(N)$ itself is known (Hiraoka & Shirai, 2017; Hino & Kanazawa, 2019); the proposition only makes the constant explicit. $\square$

Numerically, the integral equals 0.791, 3.92, 11.9, 30.2, 70.1 at $N = 16, 32, 64, 128, 256$, against simulated means 1.91, 5.83, 15.5, 36.2, 79.3 under i.i.d. Uniform $[0, 1]$ weights (Figure 4, left). Over $N \in [16, 256]$ both grow faster than linearly (log-log slopes 1.55 and 1.31), a pre-asymptotic effect of the $-\sqrt{3N}$ term; both are $\Theta(N)$ as $N \rightarrow \infty$.

A.4 DIRECTED FLAG COMPLEXES ON CAUSAL ATTENTION

Proposition 3 (Directed flag collapse). *Let A be lower-triangular with non-negative entries (self-loops are ignored). Let $\text{DFl}(A)$ be the filtered directed flag complex (Lütgehetmann et al., 2020) of the digraph with an edge $i \rightarrow j$ for every $j < i$ (a transitive tournament) and weight $1 - A_{ij}$, and let $\text{VR}(D)$ be as above with $D = 1 - \max(A, A^\top)$. Then $\text{DFl}(A)$ and $\text{VR}(D)$ are isomorphic filtered complexes, so $\text{dgm}_k(\text{DFl}(A)) = \text{dgm}_k(\text{VR}(D))$ for all k . The same holds for any weighting $w(i \rightarrow j) = \phi(A_{ij})$ with $D_{ij} = \phi(\max(A_{ij}, A_{ji}))$ and ϕ non-increasing.*

Proof. A directed k -simplex is an ordered tuple $(v_0, \dots, v_k)$ with an edge $v_a \rightarrow v_b$ for all $a < b$. Edges exist only from later to earlier tokens, so a vertex set S supports exactly one directed simplex, its elements in decreasing order; every other ordering uses a missing forward edge. (A missing edge is not a zero-weight edge; encoding it as weight 0 would admit all orderings, a common implementation error.) This bijection preserves faces. Its filtration value is $\max_{a < b} (1 - A_{v_a v_b})$, and for $u > v$, $D_{uv} = 1 - A_{uv}$, so the undirected value $\max_{u, v \in S} D_{uv}$ is the same number. $\square$

We verified the bijection and the filtration identity by explicit enumeration on 420 random causal matrices ($N \leq 8$, four families including tie-dense and underflowing matrices; 30,060 vertex sets; zero violations), and confirmed that the identity fails for bidirectional attention, where a set of size n supports $n!$ orderings. The collapse is a consequence of causality: for decoders, the directed flag complex adds nothing to its symmetrized counterpart. Directional features that do not come from flag complexes, such as those of Kushnareva et al. (2021) for bidirectional encoders, are outside this statement.

B IMPLEMENTATION AND PROTOCOL DETAILS

Extraction. All reported settings run in bfloat16 on Apple-silicon (MPS) with eager attention; float16 on MPS produced non-finite values and degenerate generations for Qwen models, and the extractor refuses non-finite logits. Model identifiers are in Table 7 (Hugging Face main revisions as of September 2026). For each example we run one forward pass with attentions and hidden states, average heads, and form D in float64. Ripser (Bauer, 2021) computes 0D/1D barcodes; per-head MSTs use a batched dense Prim’s algorithm (SciPy’s sparse MST drops explicit zero-weight edges, which occur when $A_{u0} \approx 1$). δ_0 uses the closed form of Corollary 2; $\min_s \delta_s$ is computed by brute force over all apexes. DEFLATED deletes token 0 and renormalizes each row over the remaining causal support. ANSWER-ONLY restricts D to the answer tokens without renormalization. The Lookback ratio of head h is the mean over answer tokens t of $\bar{a}_{\text{ctx}}/(\bar{a}_{\text{ctx}} + \bar{a}_{\text{new}})$, with $\bar{a}$ the mean attention from t to prompt tokens and to answer tokens up to t . The LLM-Check score of a head is the mean log diagonal attention (the log-determinant of the triangular attention kernel divided by N), summed over heads per layer. Log-probability features are the mean, minimum, and sum of answer-token log-probabilities, mean and maximum predictive entropy, and mean maximum softmax probability. Hidden states are mean-pooled over answer tokens at layer $\lfloor L/2 \rfloor$.

Prompts. TruthfulQA uses each model’s own format: the Qwen chat template with an explicit system turn “You are a helpful assistant.”, `<|user|>...<|end|>` for Phi-3, `<|user|>...</s>` for TinyLlama, ChatML without system turn for SmoLM, and `[INST]...[/INST]` for Mistral. HaluEval uses Knowledge: ... \nQuestion: ... \nAnswer: without a chat template. The prompt boundary is computed by tokenizing the prefix without its trailing space, which the full string merges into the first answer token. The on-policy prompt is the model’s chat template around “Answer the following question with a short phrase only, no explanation. Question: ...”.

Token 0 is whatever token the format places first (BOS for TinyLlama and Mistral, `<|user|>` for Phi-3, whose tokenizer adds no BOS, `<|im_start|>` for Qwen and SmolLM, and “Knowledge” for Qwen on HaluEval); the theorem does not require it to be a special token. On-policy answers were generated from the chat-formatted prompt with `add_special_tokens=False`, and features were extracted with the tokenizer’s default special tokens; the two differ only for TinyLlama, whose extraction input starts with a BOS token.

Sensitivity to inference precision. An earlier run of five settings in float16 lets us measure how much each bank depends on numerical precision (Table 5). Switching to bfloat16 changes the AUC of head-averaged topological banks by up to 0.058 (TQA Qw3B; at most 0.005 elsewhere), and of sink features by up to 0.003, but that of non-topological banks by at most 0.001. Short 1D bars live near the scale of δ_0 and are sensitive to small perturbations of attention, which is one more reason to report the precision alongside any attention-topology result.

Table 5: **AUC change from float16 to bfloat16** (same prompts and protocol; HaluEval Qwen2.5-3B also changed its prompt boundary, see above).

Setting	Sink	0D	1D	Defl.	0D _h	LogP	Lookb.	Hidden	Non-topo.
TQA Qw3B	−0.003	−0.018	+0.058	−0.001	−0.001	−0.001	0.000	0.000	−0.001
TQA Phi-3	−0.001	−0.001	−0.003	0.000	−0.001	0.000	0.000	0.000	0.000
TQA TinyLL	0.000	0.000	−0.001	−0.001	0.000	0.000	0.000	0.000	0.000
TQA SmolLM	0.000	−0.001	−0.001	+0.001	+0.002	0.000	0.000	0.000	0.000
HE Qw3B	+0.001	−0.005	+0.001	+0.002	−	−0.001	−	0.000	−

On-policy benchmark. Table 6 summarizes the generations each model produced and how they were graded.

Table 6: **TriviaQA on-policy benchmark.** 2,000 validation questions sampled with seed 0; greedy decoding, first line of the answer, at most 24 new tokens. An answer is correct if any normalized gold alias (lower-cased, punctuation and articles removed) occurs in the normalized answer as a whole-word span. Empty answers are dropped.

Model	questions	accuracy	rows used	mean answer tokens
Qwen2.5-3B	2,000	42.4%	2,000	5.5
Qwen2.5-1.5B	2,000	31.4%	2,000	5.8
Phi-3-mini	2,000	56.2%	2,000	6.0
TinyLlama-1.1B	2,000	26.4%	2,000	18.9

Statistics. Bootstrap resamples are drawn over questions (both rows of a pair move together). TOST power at margin m assumes a true difference of 0 and a normal sampling distribution with the bootstrap SE: $2\Phi(m/SE - z_{0.95}) - 1$. Repeated-CV predictions (seeds 42, 43, 44) are averaged before bootstrapping (seed 0), so the intervals reflect sampling of questions for a fixed, split-averaged predictor and omit refitting variance. The seed-to-seed SD of the pooled AUC is reported separately (Appendix E); it is at most 0.008 and exceeds 0.004 in 17 of the bank-setting cells. The C rule depends on bank width, so the same bank can be fitted with $C=1$ for a shallow model and with tuned C for a deeper one. Late fusion fits the second-level model on first-level out-of-fold scores, as in standard stacking; its training rows’ first-level scores come from models that saw the corresponding test questions, which is a mild, standard optimism shared by both arms of the comparison.

C ADDITIONAL ANALYSES

Models. Table 7 lists the audited architectures. All models use grouped or multi-head attention; we average attention over query heads.

Table 7: Audited models.

Model	params	layers	heads	KV heads	hidden
Qwen/Qwen2.5-3B-Instruct	3.09B	36	16	2	2048
Qwen/Qwen2.5-1.5B-Instruct	1.54B	28	12	2	1536
microsoft/Phi-3-mini-4k-instruct	3.82B	32	32	32	3072
TinyLlama/TinyLlama-1.1B-Chat-v1.0	1.10B	22	32	4	2048
HuggingFaceTB/SmolLM-1.7B-Instruct	1.71B	24	32	32	2048
mistralai/Mistral-7B-Instruct-v0.2	7.24B	32	32	8	4096

Length scaling of 1D persistence on real attention. Table 8 regresses log total 1D persistence (summed over layers) on log N with a class fixed effect, on rows with $P_1 > 0$, so that the exponent is not driven by a length difference between correct and hallucinated answers. Exponents range from 0.03 to 1.39, against 1.31 for i.i.d. uniform weights and 1.27 for random causal attention without a sink over the same range of N . 10 of 11 settings grow significantly more slowly than the i.i.d. reference (not HE Qw3B). The exponent exceeds 1 in HE Qw3B, HE Qw1.5B, so growth on real attention is not uniformly sub-linear. Summed over layers, almost every example has some 1D bar, because every model has some layers that are not coned; within an example, the coned layers contribute nothing to P_1 whatever the length (Table 1).

Table 8: Length scaling of 1D persistence. OLS of log P_1 on log N and the class label; rows with $P_1 = 0$ are excluded and counted in the first column.

Setting	rows with $P_1 > 0$	$\hat{\alpha}$ (class FE)	95% CI	R^2	Spearman(P_1, N)
TQA Qw3B	100.0%	0.25	[0.23, 0.27]	0.27	0.57
TQA Qw1.5B	100.0%	0.57	[0.52, 0.61]	0.29	0.51
TQA Phi-3	99.9%	0.93	[0.82, 1.04]	0.15	0.39
TQA TinyLL	100.0%	0.32	[0.28, 0.36]	0.14	0.28
TQA SmolLM	100.0%	0.61	[0.54, 0.69]	0.13	0.36
HE Qw3B	100.0%	1.39	[1.31, 1.47]	0.36	0.60
HE Qw1.5B	100.0%	1.10	[1.07, 1.13]	0.84	0.91
TrQA Qw3B	100.0%	0.72	[0.68, 0.75]	0.46	0.68
TrQA Qw1.5B	100.0%	0.75	[0.72, 0.78]	0.59	0.68
TrQA Phi-3	100.0%	0.70	[0.65, 0.75]	0.29	0.49
TrQA TinyLL	100.0%	0.03	[0.02, 0.03]	0.03	0.15
i.i.d. uniform weights (simulation)	100%	1.31	—	—	—
random causal attention, no sink (simulation)	100%	1.27	—	—	—

Sequence lengths. Table 9 gives class-conditional lengths. TruthfulQA is length-balanced by construction; HaluEval’s hallucinated answers are systematically longer; on-policy, incorrect answers differ in length from correct ones because they are the model’s own.

Table 9: Class-conditional sequence length (mean $\pm$ SD, full sequence) and mean answer length.

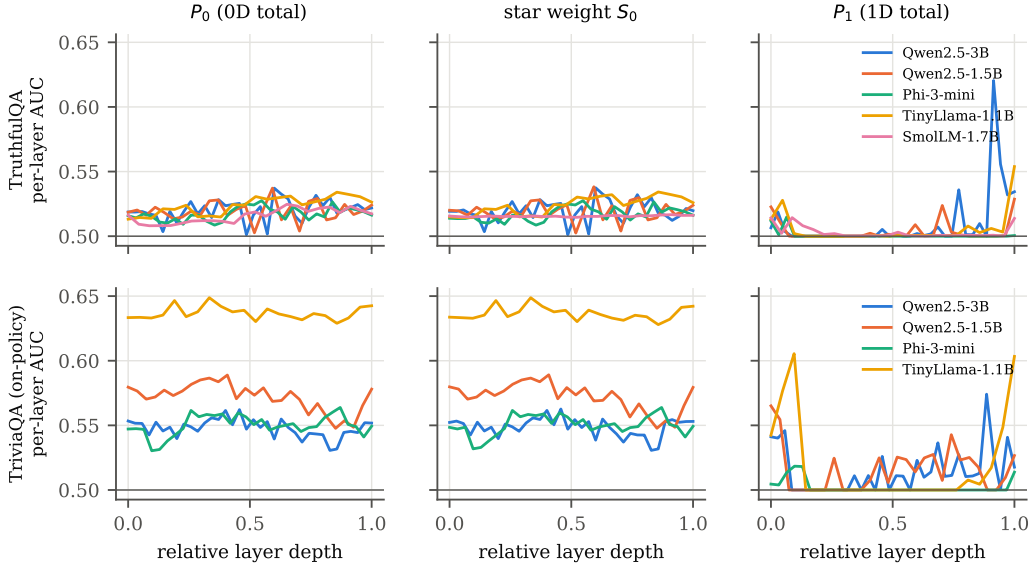
Setting	N (correct)	N (hallucinated)	answer tokens (c./h.)	point-biserial r	Welch p
TQA Qw3B	44.4 $\pm$ 7.9	44.0 $\pm$ 8.0	12.6 / 12.1	-0.030	0.22
TQA Qw1.5B	44.4 $\pm$ 7.9	44.0 $\pm$ 8.0	12.6 / 12.1	-0.030	0.22
TQA Phi-3	29.2 $\pm$ 8.9	28.7 $\pm$ 8.8	11.8 / 11.4	-0.024	0.32
TQA TinyLL	42.5 $\pm$ 9.0	42.1 $\pm$ 8.9	12.1 / 11.7	-0.024	0.32
TQA SmolLM	32.6 $\pm$ 8.2	32.1 $\pm$ 8.2	10.6 / 10.2	-0.026	0.29
HE Qw3B	121.2 $\pm$ 39.7	132.7 $\pm$ 41.2	3.8 / 15.4	+0.141	$< 10^{-9}$
HE Qw1.5B	122.2 $\pm$ 39.5	133.9 $\pm$ 41.3	3.6 / 15.3	+0.144	$< 10^{-5}$
TrQA Qw3B	68.0 $\pm$ 10.4	69.0 $\pm$ 9.7	5.3 / 5.7	+0.052	0.022
TrQA Qw1.5B	67.5 $\pm$ 10.3	69.4 $\pm$ 10.0	5.4 / 5.9	+0.086	0.00013
TrQA Phi-3	45.6 $\pm$ 11.3	46.9 $\pm$ 11.2	5.6 / 6.5	+0.057	0.01
TrQA TinyLL	67.3 $\pm$ 12.5	73.7 $\pm$ 14.9	17.0 / 19.6	+0.196	$< 10^{-20}$

Margin sensitivity. Table 10 reports TOST p -values at four equivalence margins for the four comparisons of Figure 5; bold marks $p < 0.05$ (equivalence).

Table 10: **TOST p -values by margin m** . NT: all non-topological features.

Setting m	Sink vs. 0D				0D Sink				1D 0D				0D NT			
	.010	.015	.020	.025	.010	.015	.020	.025	.010	.015	.020	.025	.010	.015	.020	.025
TQA Qw3B	.029	.001	<.001	<.001	.040	.001	<.001	<.001	1.000	.995	.972	.882	<.001	<.001	<.001	<.001
TQA Qw1.5B	.109	.016	<.001	<.001	.692	.276	.040	.002	.311	.034	<.001	<.001	<.001	<.001	<.001	<.001
TQA Phi-3	.138	.013	<.001	<.001	.112	.003	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001
TQA TinyLL	.520	.234	.064	.012	.962	.809	.483	.169	.210	.021	<.001	<.001	<.001	<.001	<.001	<.001
TQA SmolLM	.939	.865	.748	.602	1.000	.996	.960	.805	.179	.036	.003	<.001	<.001	<.001	<.001	<.001
HE Qw3B	.910	.676	.351	.110	1.000	1.000	.982	.804	.906	.551	.148	.012	<.001	<.001	<.001	<.001
HE Qw1.5B	.993	.970	.896	.755	.679	.263	.041	.002	.980	.918	.786	.569	<.001	<.001	<.001	<.001
TrQA Qw3B	.792	.522	.239	.069	.982	.866	.561	.219	.106	.015	.002	<.001	<.001	<.001	<.001	<.001
TrQA Qw1.5B	.431	.176	.050	.010	.528	.211	.044	.005	.079	.011	<.001	<.001	<.001	<.001	<.001	<.001
TrQA Phi-3	.047	.002	<.001	<.001	.114	.005	<.001	<.001	.086	.003	<.001	<.001	<.001	<.001	<.001	<.001
TrQA TinyLL	.091	.021	.003	<.001	.071	.005	<.001	<.001	.083	.013	.002	<.001	<.001	<.001	<.001	<.001

Per-layer signal. Figure 2 shows single-feature AUCs by depth. P_0 and the star weight trace nearly identical curves in every model with a sink, while P_1 carries signal only in the layers that are not coned. These AUCs are orientation-free ($\max(\text{AUC}, 1 - \text{AUC})$) and therefore slightly optimistic; they are descriptive, not a detector.

Figure 2: **Single-feature AUC by layer.**

Worked example. Figure 3 shows TruthfulQA question 67 with its incorrect answer (“Yes, vampires are real”) under TinyLlama-1.1B ($N=29$). In layer 5 the model sends 0.85 of its attention to token 0 on average, $\delta_0 = 0.000$, and accordingly $P_0 = 4.225$ equals the star weight $S_0 = 4.225$ and there are 0 H_1 bars. In layer 21 sink attention drops to 0.53 and $\delta_0 = 0.281$; now $P_0 = 12.287$ falls below $S_0 = 13.043$ (within the bound $(N-1)\delta_0 = 7.87$), and 10 H_1 bars appear, the longest of length $0.172 \leq \delta_0$. The theorem bounds loops but does not force them: in layers 2 and 11–18 of the same example δ_0 lies between 0.11 and 0.52, yet H_1 is empty.

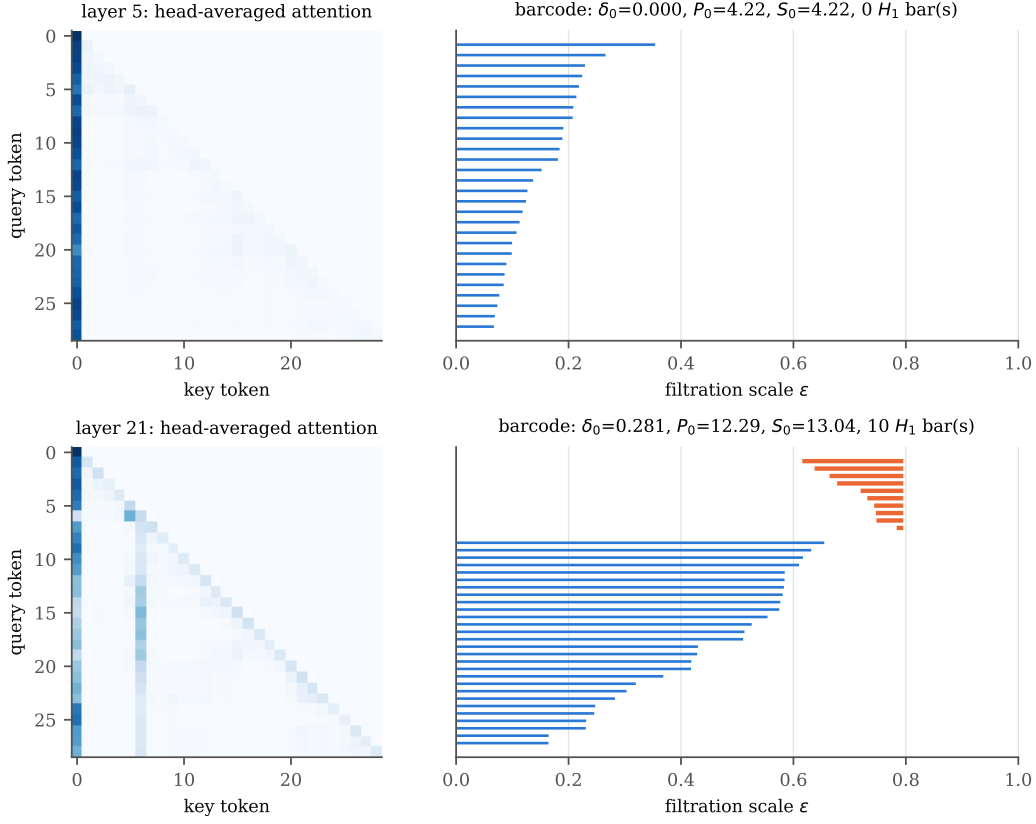


Figure 3: **A coned and a non-coned layer of the same example** (TinyLlama-1.1B, TruthfulQA). Left: head-averaged attention; the bright first column is the sink. Right: 0D bars (blue) and 1D bars (orange) of the Vietoris–Rips filtration of $D = 1 - \max(A, A^\top)$.

D ONE-DIMENSIONAL PERSISTENCE

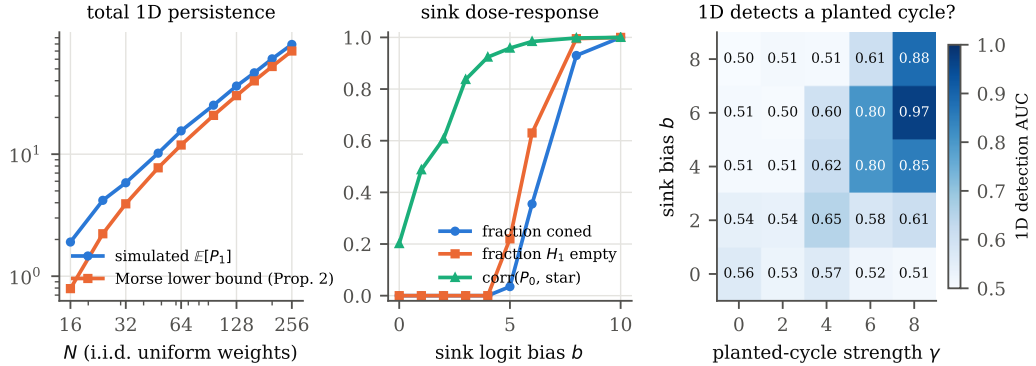


Figure 4: **Synthetic checks.** *Left:* Proposition 2’s lower bound against the simulated mean total 1D persistence under i.i.d. Uniform[0, 1] weights (20 draws per $N \leq 128$, 8 above). *Middle:* random causal attention ($N=40$, 200 draws per point) with a logit bias b on token 0: as the sink strengthens, graphs become exactly coned, H_1 empties, and P_0 becomes a function of the star weight. *Right:* AUC of 1D total persistence for detecting a planted 6-cycle of logit strength γ (150 draws per class). Without a sink ($b=0$) the planted cycle is lost among background cycles; under a strong sink ($b=8$) it is invisible unless it is strong enough to break the cone.

Under i.i.d. weights, 1D persistence grows faster than linearly over the lengths that matter here and linearly asymptotically (Proposition 2; Figure 4, left). Real attention behaves very differently: most graphs have *no* 1D bars at all (Table 1), and the bars that exist are confined to graphs with a non-zero coning defect. Figure 4 (middle) reproduces this transition in a controlled setting: raising the sink’s logit bias moves random causal attention from the i.i.d.-like regime to exact coning, with H_1 emptying in lockstep.

This explains a recurring empirical observation, that 1D features add little to 0D. At full sample size, adding length-normalized 1D to 0D changes AUC by -0.008 to $+0.033$ and is equivalent within ± 0.015 in 7 of 10 sink settings; the exceptions are TQA Qw3B, HE Qw3B and HE Qw1.5B, where adding 1D helps significantly. The TruthfulQA Qwen2.5-3B gain is also the result that is most sensitive to inference precision (Appendix B). The mechanism is not that loops are uninformative in principle. In a strongly sinked model, a loop can only exist where some token pays another token more attention than one of the two pays to the sink, and its lifetime is bounded by that excess. Figure 4 (right) makes this concrete. A planted 6-cycle that H_1 detects easily under a moderate sink ($b=6, \gamma=8$: AUC 0.97) is invisible under a strong one ($b=8, \gamma \leq 4$: AUC ≤ 0.51) until it is strong enough to break the cone. Without any sink ($b=0$) H_1 also misses it (AUC ≤ 0.57), because random attention already has many background cycles; 1D persistence is informative only in between.

E FULL RESULTS

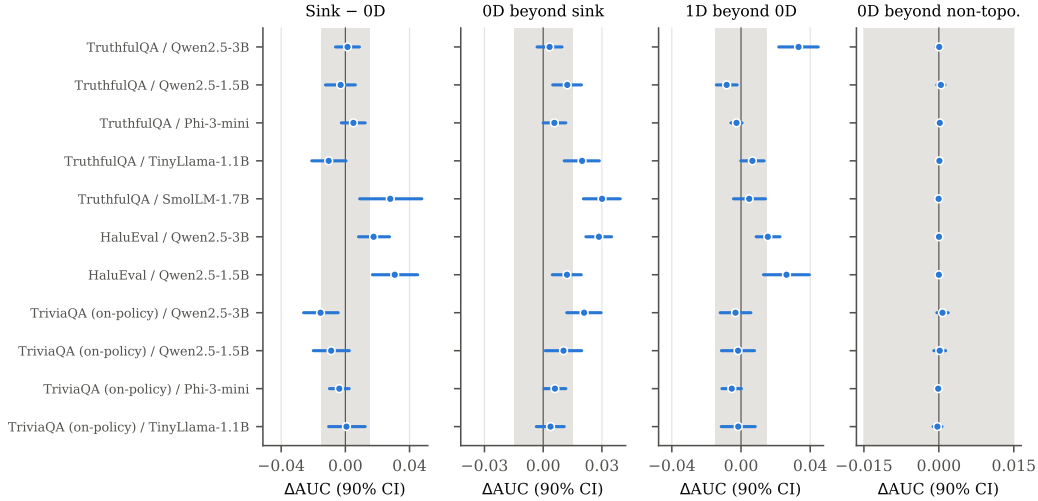


Figure 5: **Four comparisons across all settings** (ΔAUC with 90% cluster-bootstrap intervals; shaded band: ± 0.015). Sink features nearly reproduce 0D; 0D adds little beyond them; 1D adds little beyond 0D except in three settings; and 0D adds nothing beyond non-topological probes under early fusion (late fusion in Table 3).

Setting	Bank	dim	AUC	seed SD	pair acc.
TQA Qw3B	LEN	2	0.529	0.0003	0.567
TQA Qw3B	LEX	–	0.820	0.0014	0.829
TQA Qw3B	LOGPROB	6	0.630	0.0006	0.694
TQA Qw3B	LLMCHECK	36	0.693	0.0011	0.764
TQA Qw3B	ROWSTAT	72	0.750	0.0003	0.823
TQA Qw3B	SINK	72	0.737	0.0022	0.767
TQA Qw3B	0D	72	0.736	0.0013	0.778
TQA Qw3B	LEGACY MSTPROXY	36	0.713	0.0008	0.788
TQA Qw3B	1D	144	0.660	0.0001	0.674
TQA Qw3B	DEFL	216	0.715	0.0028	0.753
TQA Qw3B	ANS	216	0.748	0.0017	0.749
TQA Qw3B	PH SINK	576	0.851	0.0015	0.870

	Setting	Bank	dim	AUC	seed SD	pair acc.
1134						
1135						
1136	TQA Qw3B	PH 0DTOT	576	0.842	0.0011	0.869
1137	TQA Qw3B	PH 0D	1152	0.853	0.0013	0.862
1138	TQA Qw3B	PH ENT	576	0.857	0.0013	0.901
1139	TQA Qw3B	LOOKBACK	576	0.897	0.0022	0.908
1140	TQA Qw3B	HIDDEN	2048	0.913	0.0008	0.918
1141	TQA Qw3B	NONTOPO	2738	0.926	0.0006	0.936
1142	TQA Qw3B	0D+1D	216	0.769	0.0004	0.791
1143	TQA Qw3B	0D+SINK	144	0.741	0.0015	0.776
1144	TQA Qw3B	HIDDEN+0D	2120	0.913	0.0007	0.920
1145	TQA Qw3B	LEX+0D	–	0.840	0.0015	0.848
1146	TQA Qw3B	LEX+DEFL	–	0.830	0.0004	0.854
1147	TQA Qw3B	LEX+SINK	–	0.843	0.0020	0.851
1148	TQA Qw3B	LOGPROB+0D	78	0.776	0.0022	0.810
1149	TQA Qw3B	NONTOPO+0D	2810	0.926	0.0004	0.936
1150	TQA Qw3B	NONTOPO+DEFL	2954	0.928	0.0006	0.933
1151	TQA Qw3B	PH SINK+PH 0DTOT	1152	0.860	0.0022	0.892
1152	TQA Qw3B	SINK+1D	216	0.766	0.0007	0.787
1153	TQA Qw3B	SINK+ANS	288	0.785	0.0031	0.797
1154	TQA Qw3B	SINK+DEFL	288	0.766	0.0024	0.799
1155	TQA Qw1.5B	LEN	2	0.529	0.0003	0.567
1156	TQA Qw1.5B	LEX	–	0.820	0.0014	0.829
1157	TQA Qw1.5B	LOGPROB	6	0.638	0.0007	0.682
1158	TQA Qw1.5B	LLMCHECK	28	0.666	0.0005	0.743
1159	TQA Qw1.5B	ROWSTAT	56	0.738	0.0014	0.815
1160	TQA Qw1.5B	SINK	56	0.723	0.0019	0.771
1161	TQA Qw1.5B	0D	56	0.726	0.0007	0.767
1162	TQA Qw1.5B	LEGACY MSTPROXY	28	0.698	0.0010	0.786
1163	TQA Qw1.5B	1D	112	0.586	0.0014	0.644
1164	TQA Qw1.5B	DEFL	168	0.703	0.0042	0.755
1165	TQA Qw1.5B	ANS	168	0.697	0.0040	0.700
1166	TQA Qw1.5B	PH SINK	336	0.830	0.0008	0.851
1167	TQA Qw1.5B	PH 0DTOT	336	0.819	0.0011	0.838
1168	TQA Qw1.5B	PH 0D	672	0.838	0.0012	0.847
1169	TQA Qw1.5B	PH ENT	336	0.833	0.0025	0.852
1170	TQA Qw1.5B	LOOKBACK	336	0.851	0.0020	0.856
1171	TQA Qw1.5B	HIDDEN	1536	0.916	0.0012	0.923
1172	TQA Qw1.5B	NONTOPO	1962	0.917	0.0013	0.928
1173	TQA Qw1.5B	0D+1D	168	0.718	0.0013	0.767
1174	TQA Qw1.5B	0D+SINK	112	0.735	0.0012	0.766
1175	TQA Qw1.5B	HIDDEN+0D	1592	0.916	0.0014	0.924
1176	TQA Qw1.5B	LEX+0D	–	0.835	0.0017	0.841
1177	TQA Qw1.5B	LEX+DEFL	–	0.825	0.0019	0.836
1178	TQA Qw1.5B	LEX+SINK	–	0.833	0.0008	0.846
1179	TQA Qw1.5B	LOGPROB+0D	62	0.740	0.0013	0.797
1180	TQA Qw1.5B	NONTOPO+0D	2018	0.917	0.0014	0.927
1181	TQA Qw1.5B	NONTOPO+DEFL	2130	0.917	0.0014	0.923
1182	TQA Qw1.5B	PH SINK+PH 0DTOT	672	0.841	0.0013	0.867
1183	TQA Qw1.5B	SINK+1D	168	0.715	0.0017	0.769
1184	TQA Qw1.5B	SINK+ANS	224	0.761	0.0027	0.753
1185	TQA Qw1.5B	SINK+DEFL	224	0.757	0.0028	0.802
1186	TQA Phi-3	LEN	2	0.522	0.0004	0.562
1187	TQA Phi-3	LEX	–	0.820	0.0014	0.829
	TQA Phi-3	LOGPROB	6	0.578	0.0016	0.597
	TQA Phi-3	LLMCHECK	32	0.653	0.0008	0.755
	TQA Phi-3	ROWSTAT	64	0.661	0.0011	0.738
	TQA Phi-3	SINK	64	0.651	0.0006	0.700
	TQA Phi-3	0D	64	0.646	0.0011	0.709
	TQA Phi-3	LEGACY MSTPROXY	32	0.613	0.0008	0.694
	TQA Phi-3	1D	128	0.526	0.0030	0.551
	TQA Phi-3	DEFL	192	0.639	0.0011	0.684
	TQA Phi-3	ANS	192	0.629	0.0042	0.651
	TQA Phi-3	PH SINK	1024	0.871	0.0008	0.901
	TQA Phi-3	PH 0DTOT	1024	0.845	0.0014	0.882

	Setting	Bank	dim	AUC	seed SD	pair acc.
1188						
1189						
1190	TQA Phi-3	PH 0D	2048	0.868	0.0007	0.881
1191	TQA Phi-3	PH ENT	1024	0.864	0.0010	0.895
1192	TQA Phi-3	LOOKBACK	1024	0.892	0.0004	0.902
1193	TQA Phi-3	HIDDEN	3072	0.933	0.0007	0.945
1194	TQA Phi-3	NONTOPO	4198	0.936	0.0006	0.942
1195	TQA Phi-3	0D+1D	192	0.643	0.0014	0.705
1196	TQA Phi-3	0D+SINK	128	0.657	0.0007	0.704
1197	TQA Phi-3	HIDDEN+0D	3136	0.933	0.0008	0.944
1198	TQA Phi-3	LEX+0D	–	0.821	0.0003	0.842
1199	TQA Phi-3	LEX+DEFL	–	0.802	0.0001	0.827
1200	TQA Phi-3	LEX+SINK	–	0.820	0.0007	0.834
1201	TQA Phi-3	LOGPROB+0D	70	0.655	0.0011	0.694
1202	TQA Phi-3	NONTOPO+0D	4262	0.936	0.0006	0.942
1203	TQA Phi-3	NONTOPO+DEFL	4390	0.937	0.0007	0.939
1204	TQA Phi-3	PH SINK+PH 0DTOT	2048	0.871	0.0007	0.905
1205	TQA Phi-3	SINK+1D	192	0.651	0.0011	0.696
1206	TQA Phi-3	SINK+ANS	256	0.675	0.0015	0.698
1207	TQA Phi-3	SINK+DEFL	256	0.684	0.0023	0.718
1208	TQA TinyLl.	LEN	2	0.523	0.0005	0.556
1209	TQA TinyLl.	LEX	–	0.820	0.0014	0.829
1210	TQA TinyLl.	LOGPROB	6	0.637	0.0010	0.645
1211	TQA TinyLl.	LLMCHECK	22	0.614	0.0005	0.705
1212	TQA TinyLl.	ROWSTAT	44	0.661	0.0015	0.749
1213	TQA TinyLl.	SINK	44	0.643	0.0006	0.718
1214	TQA TinyLl.	0D	44	0.654	0.0011	0.729
1215	TQA TinyLl.	LEGACY MSTPROXY	22	0.627	0.0009	0.741
1216	TQA TinyLl.	1D	88	0.592	0.0006	0.673
1217	TQA TinyLl.	DEFL	132	0.651	0.0019	0.705
1218	TQA TinyLl.	ANS	132	0.644	0.0049	0.678
1219	TQA TinyLl.	PH SINK	704	0.808	0.0015	0.841
1220	TQA TinyLl.	PH 0DTOT	704	0.804	0.0021	0.830
1221	TQA TinyLl.	PH 0D	1408	0.806	0.0015	0.820
1222	TQA TinyLl.	PH ENT	704	0.817	0.0008	0.838
1223	TQA TinyLl.	LOOKBACK	704	0.850	0.0028	0.846
1224	TQA TinyLl.	HIDDEN	2048	0.883	0.0013	0.879
1225	TQA TinyLl.	NONTOPO	2824	0.885	0.0011	0.887
1226	TQA TinyLl.	0D+1D	132	0.660	0.0010	0.744
1227	TQA TinyLl.	0D+SINK	88	0.663	0.0004	0.728
1228	TQA TinyLl.	HIDDEN+0D	2092	0.883	0.0014	0.880
1229	TQA TinyLl.	LEX+0D	–	0.830	0.0012	0.836
1230	TQA TinyLl.	LEX+DEFL	–	0.826	0.0012	0.846
1231	TQA TinyLl.	LEX+SINK	–	0.831	0.0011	0.832
1232	TQA TinyLl.	LOGPROB+0D	50	0.683	0.0010	0.711
1233	TQA TinyLl.	NONTOPO+0D	2868	0.885	0.0011	0.889
1234	TQA TinyLl.	NONTOPO+DEFL	2956	0.884	0.0010	0.885
1235	TQA TinyLl.	PH SINK+PH 0DTOT	1408	0.810	0.0031	0.841
1236	TQA TinyLl.	SINK+1D	132	0.651	0.0016	0.729
1237	TQA TinyLl.	SINK+ANS	176	0.685	0.0041	0.720
1238	TQA TinyLl.	SINK+DEFL	176	0.688	0.0015	0.749
1239	TQA SmolLM	LEN	2	0.524	0.0003	0.554
1240	TQA SmolLM	LEX	–	0.820	0.0014	0.829
1241	TQA SmolLM	LOGPROB	6	0.588	0.0010	0.592
	TQA SmolLM	LLMCHECK	24	0.604	0.0004	0.684
	TQA SmolLM	ROWSTAT	48	0.673	0.0014	0.741
	TQA SmolLM	SINK	48	0.611	0.0019	0.628
	TQA SmolLM	0D	48	0.583	0.0029	0.656
	TQA SmolLM	LEGACY MSTPROXY	24	0.585	0.0011	0.674
	TQA SmolLM	1D	96	0.543	0.0015	0.581
	TQA SmolLM	DEFL	144	0.594	0.0023	0.650
	TQA SmolLM	ANS	144	0.628	0.0041	0.630
	TQA SmolLM	PH SINK	768	0.720	0.0008	0.775
	TQA SmolLM	PH 0DTOT	768	0.819	0.0019	0.845
	TQA SmolLM	PH 0D	1536	0.800	0.0012	0.818

	Setting	Bank	dim	AUC	seed SD	pair acc.
1242	TQA SmolLM	PH ENT	768	0.834	0.0007	0.841
1243	TQA SmolLM	LOOKBACK	768	0.863	0.0017	0.864
1244	TQA SmolLM	HIDDEN	2048	0.894	0.0022	0.892
1245	TQA SmolLM	NONTOPO	2894	0.899	0.0015	0.903
1246	TQA SmolLM	0D+1D	144	0.588	0.0019	0.643
1247	TQA SmolLM	0D+SINK	96	0.641	0.0026	0.662
1248	TQA SmolLM	HIDDEN+0D	2096	0.894	0.0020	0.891
1249	TQA SmolLM	LEX+0D	–	0.827	0.0008	0.838
1250	TQA SmolLM	LEX+DEFL	–	0.822	0.0001	0.845
1251	TQA SmolLM	LEX+SINK	–	0.819	0.0015	0.826
1252	TQA SmolLM	LOGPROB+0D	54	0.613	0.0009	0.630
1253	TQA SmolLM	NONTOPO+0D	2942	0.899	0.0013	0.902
1254	TQA SmolLM	NONTOPO+DEFL	3038	0.899	0.0012	0.900
1255	TQA SmolLM	PH SINK+PH 0DTOT	1536	0.826	0.0023	0.854
1256	TQA SmolLM	SINK+1D	144	0.629	0.0013	0.650
1257	TQA SmolLM	SINK+ANS	192	0.646	0.0021	0.651
1258	TQA SmolLM	SINK+DEFL	192	0.652	0.0022	0.681
1259	HE Qw3B	LEN	2	0.962	0.0001	0.970
1260	HE Qw3B	LEX	–	0.981	0.0004	0.977
1261	HE Qw3B	LOGPROB	6	0.993	0.0001	0.991
1262	HE Qw3B	LLMCHECK	36	0.745	0.0004	0.956
1263	HE Qw3B	ROWSTAT	72	0.829	0.0002	0.976
1264	HE Qw3B	SINK	72	0.829	0.0008	0.930
1265	HE Qw3B	0D	72	0.811	0.0006	0.927
1266	HE Qw3B	LEGACY MSTPROXY	36	0.731	0.0002	0.960
1267	HE Qw3B	1D	144	0.710	0.0013	0.891
1268	HE Qw3B	DEFL	216	0.834	0.0009	0.959
1269	HE Qw3B	ANS	216	0.978	0.0009	0.980
1270	HE Qw3B	PH SINK	576	0.958	0.0007	0.999
1271	HE Qw3B	PH 0DTOT	576	0.958	0.0005	0.995
1272	HE Qw3B	PH 0D	1152	0.971	0.0006	0.988
1273	HE Qw3B	PH ENT	576	0.964	0.0005	0.993
1274	HE Qw3B	LOOKBACK	576	0.998	0.0001	0.998
1275	HE Qw3B	HIDDEN	2048	1.000	0.0000	1.000
1276	HE Qw3B	NONTOPO	2738	1.000	0.0002	1.000
1277	HE Qw3B	0D+1D	216	0.826	0.0013	0.945
1278	HE Qw3B	0D+SINK	144	0.857	0.0010	0.948
1279	HE Qw3B	HIDDEN+0D	2120	1.000	0.0000	1.000
1280	HE Qw3B	LEX+0D	–	0.979	0.0001	0.983
1281	HE Qw3B	LEX+DEFL	–	0.971	0.0003	0.975
1282	HE Qw3B	LEX+SINK	–	0.980	0.0003	0.981
1283	HE Qw3B	LOGPROB+0D	78	0.993	0.0003	0.992
1284	HE Qw3B	NONTOPO+0D	2810	1.000	0.0002	1.000
1285	HE Qw3B	NONTOPO+DEFL	2954	1.000	0.0002	1.000
1286	HE Qw3B	PH SINK+PH 0DTOT	1152	0.962	0.0006	0.997
1287	HE Qw3B	SINK+1D	216	0.843	0.0011	0.938
1288	HE Qw3B	SINK+ANS	288	0.978	0.0003	0.982
1289	HE Qw3B	SINK+DEFL	288	0.887	0.0020	0.957
1290	HE Qw1.5B	LEN	2	0.967	0.0002	0.976
1291	HE Qw1.5B	LEX	–	0.979	0.0004	0.980
1292	HE Qw1.5B	LOGPROB	6	0.995	0.0002	0.988
1293	HE Qw1.5B	LLMCHECK	28	0.791	0.0005	0.978
1294	HE Qw1.5B	ROWSTAT	56	0.788	0.0017	0.984
1295	HE Qw1.5B	SINK	56	0.820	0.0015	0.934
	HE Qw1.5B	0D	56	0.789	0.0014	0.918
	HE Qw1.5B	LEGACY MSTPROXY	28	0.719	0.0003	0.976
	HE Qw1.5B	1D	112	0.747	0.0006	0.878
	HE Qw1.5B	DEFL	168	0.825	0.0021	0.952
	HE Qw1.5B	ANS	168	0.972	0.0006	0.978
	HE Qw1.5B	PH SINK	336	0.922	0.0007	0.998
	HE Qw1.5B	PH 0DTOT	336	0.921	0.0007	0.996
	HE Qw1.5B	PH 0D	672	0.949	0.0027	0.974
	HE Qw1.5B	PH ENT	336	0.933	0.0011	0.996

	Setting	Bank	dim	AUC	seed SD	pair acc.
1296						
1297						
1298	HE Qw1.5B	LOOKBACK	336	0.999	0.0002	0.998
1299	HE Qw1.5B	HIDDEN	1536	1.000	0.0001	1.000
1300	HE Qw1.5B	NONTOPO	1962	1.000	0.0002	1.000
1301	HE Qw1.5B	0D+1D	168	0.815	0.0032	0.922
1302	HE Qw1.5B	0D+SINK	112	0.832	0.0015	0.924
1303	HE Qw1.5B	HIDDEN+0D	1592	1.000	0.0001	1.000
1304	HE Qw1.5B	LEX+0D	–	0.971	0.0004	0.986
1305	HE Qw1.5B	LEX+DEFL	–	0.956	0.0002	0.986
1306	HE Qw1.5B	LEX+SINK	–	0.968	0.0005	0.984
1307	HE Qw1.5B	LOGPROB+0D	62	0.994	0.0002	0.990
1308	HE Qw1.5B	NONTOPO+0D	2018	1.000	0.0000	1.000
1309	HE Qw1.5B	NONTOPO+DEFL	2130	1.000	0.0001	1.000
1310	HE Qw1.5B	PH SINK+PH 0DTOT	672	0.931	0.0024	0.998
1311	HE Qw1.5B	SINK+1D	168	0.826	0.0035	0.940
1312	HE Qw1.5B	SINK+ANS	224	0.975	0.0015	0.990
1313	HE Qw1.5B	SINK+DEFL	224	0.872	0.0028	0.962
1314	TrQA Qw3B	LEN	2	0.569	0.0014	–
1315	TrQA Qw3B	LEX	–	0.596	0.0039	–
1316	TrQA Qw3B	LOGPROB	6	0.839	0.0002	–
1317	TrQA Qw3B	LLMCHECK	36	0.688	0.0012	–
1318	TrQA Qw3B	ROWSTAT	72	0.724	0.0046	–
1319	TrQA Qw3B	SINK	72	0.717	0.0038	–
1320	TrQA Qw3B	0D	72	0.732	0.0038	–
1321	TrQA Qw3B	LEGACY MSTPROXY	36	0.691	0.0033	–
1322	TrQA Qw3B	1D	144	0.601	0.0031	–
1323	TrQA Qw3B	DEFL	216	0.684	0.0043	–
1324	TrQA Qw3B	ANS	216	0.719	0.0020	–
1325	TrQA Qw3B	PH SINK	576	0.820	0.0021	–
1326	TrQA Qw3B	PH 0DTOT	576	0.807	0.0012	–
1327	TrQA Qw3B	PH 0D	1152	0.807	0.0017	–
1328	TrQA Qw3B	PH ENT	576	0.810	0.0007	–
1329	TrQA Qw3B	LOOKBACK	576	0.861	0.0013	–
1330	TrQA Qw3B	HIDDEN	2048	0.841	0.0042	–
1331	TrQA Qw3B	NONTOPO	2738	0.870	0.0028	–
1332	TrQA Qw3B	0D+1D	216	0.729	0.0022	–
1333	TrQA Qw3B	0D+SINK	144	0.738	0.0050	–
1334	TrQA Qw3B	HIDDEN+0D	2120	0.843	0.0040	–
1335	TrQA Qw3B	LEX+0D	–	0.733	0.0011	–
1336	TrQA Qw3B	LEX+DEFL	–	0.690	0.0031	–
1337	TrQA Qw3B	LEX+SINK	–	0.724	0.0005	–
1338	TrQA Qw3B	LOGPROB+0D	78	0.843	0.0022	–
1339	TrQA Qw3B	NONTOPO+0D	2810	0.870	0.0027	–
1340	TrQA Qw3B	NONTOPO+DEFL	2954	0.870	0.0029	–
1341	TrQA Qw3B	PH SINK+PH 0DTOT	1152	0.824	0.0017	–
1342	TrQA Qw3B	SINK+1D	216	0.718	0.0035	–
1343	TrQA Qw3B	SINK+ANS	288	0.752	0.0010	–
1344	TrQA Qw3B	SINK+DEFL	288	0.719	0.0038	–
1345	TrQA Qw1.5B	LEN	2	0.588	0.0012	–
1346	TrQA Qw1.5B	LEX	–	0.609	0.0033	–
1347	TrQA Qw1.5B	LOGPROB	6	0.823	0.0006	–
1348	TrQA Qw1.5B	LLMCHECK	28	0.676	0.0029	–
1349	TrQA Qw1.5B	ROWSTAT	56	0.720	0.0008	–
	TrQA Qw1.5B	SINK	56	0.707	0.0000	–
	TrQA Qw1.5B	0D	56	0.716	0.0004	–
	TrQA Qw1.5B	LEGACY MSTPROXY	28	0.685	0.0014	–
	TrQA Qw1.5B	1D	112	0.574	0.0024	–
	TrQA Qw1.5B	DEFL	168	0.655	0.0016	–
	TrQA Qw1.5B	ANS	168	0.675	0.0035	–
	TrQA Qw1.5B	PH SINK	336	0.767	0.0034	–
	TrQA Qw1.5B	PH 0DTOT	336	0.759	0.0051	–
	TrQA Qw1.5B	PH 0D	672	0.768	0.0013	–
	TrQA Qw1.5B	PH ENT	336	0.768	0.0025	–
	TrQA Qw1.5B	LOOKBACK	336	0.824	0.0017	–

	Setting	Bank	dim	AUC	seed SD	pair acc.
1350						
1351						
1352	TrQA Qw1.5B	HIDDEN	1536	0.855	0.0031	—
1353	TrQA Qw1.5B	NONTOPO	1962	0.873	0.0023	—
1354	TrQA Qw1.5B	0D+1D	168	0.714	0.0015	—
1355	TrQA Qw1.5B	0D+SINK	112	0.717	0.0006	—
1356	TrQA Qw1.5B	HIDDEN+0D	1592	0.855	0.0032	—
1357	TrQA Qw1.5B	LEX+0D	—	0.721	0.0031	—
1358	TrQA Qw1.5B	LEX+DEFL	—	0.672	0.0047	—
1359	TrQA Qw1.5B	LEX+SINK	—	0.715	0.0036	—
1360	TrQA Qw1.5B	LOGPROB+0D	62	0.828	0.0007	—
1361	TrQA Qw1.5B	NONTOPO+0D	2018	0.873	0.0025	—
1362	TrQA Qw1.5B	NONTOPO+DEFL	2130	0.872	0.0024	—
1363	TrQA Qw1.5B	PH SINK+PH 0DTOT	672	0.772	0.0022	—
1364	TrQA Qw1.5B	SINK+1D	168	0.709	0.0028	—
1365	TrQA Qw1.5B	SINK+ANS	224	0.726	0.0059	—
1366	TrQA Qw1.5B	SINK+DEFL	224	0.719	0.0023	—
1367	TrQA Phi-3	LEN	2	0.581	0.0022	—
1368	TrQA Phi-3	LEX	—	0.635	0.0083	—
1369	TrQA Phi-3	LOGPROB	6	0.877	0.0002	—
1370	TrQA Phi-3	LLMCHECK	32	0.676	0.0018	—
1371	TrQA Phi-3	ROWSTAT	64	0.728	0.0006	—
1372	TrQA Phi-3	SINK	64	0.707	0.0018	—
1373	TrQA Phi-3	0D	64	0.710	0.0022	—
1374	TrQA Phi-3	LEGACY MSTPROXY	32	0.669	0.0002	—
1375	TrQA Phi-3	1D	128	0.546	0.0031	—
1376	TrQA Phi-3	DEFL	192	0.698	0.0021	—
1377	TrQA Phi-3	ANS	192	0.710	0.0037	—
1378	TrQA Phi-3	PH SINK	1024	0.826	0.0010	—
1379	TrQA Phi-3	PH 0DTOT	1024	0.810	0.0016	—
1380	TrQA Phi-3	PH 0D	2048	0.807	0.0024	—
1381	TrQA Phi-3	PH ENT	1024	0.817	0.0021	—
1382	TrQA Phi-3	LOOKBACK	1024	0.840	0.0016	—
1383	TrQA Phi-3	HIDDEN	3072	0.867	0.0035	—
1384	TrQA Phi-3	NONTOPO	4198	0.883	0.0030	—
1385	TrQA Phi-3	0D+1D	192	0.705	0.0011	—
1386	TrQA Phi-3	0D+SINK	128	0.713	0.0024	—
1387	TrQA Phi-3	HIDDEN+0D	3136	0.867	0.0037	—
1388	TrQA Phi-3	LEX+0D	—	0.721	0.0031	—
1389	TrQA Phi-3	LEX+DEFL	—	0.712	0.0022	—
1390	TrQA Phi-3	LEX+SINK	—	0.718	0.0044	—
1391	TrQA Phi-3	LOGPROB+0D	70	0.878	0.0013	—
1392	TrQA Phi-3	NONTOPO+0D	4262	0.883	0.0030	—
1393	TrQA Phi-3	NONTOPO+DEFL	4390	0.885	0.0030	—
1394	TrQA Phi-3	PH SINK+PH 0DTOT	2048	0.828	0.0009	—
1395	TrQA Phi-3	SINK+1D	192	0.699	0.0012	—
1396	TrQA Phi-3	SINK+ANS	256	0.736	0.0046	—
1397	TrQA Phi-3	SINK+DEFL	256	0.736	0.0016	—
1398	TrQA TinyLl.	LEN	2	0.631	0.0005	—
1399	TrQA TinyLl.	LEX	—	0.740	0.0003	—
1400	TrQA TinyLl.	LOGPROB	6	0.724	0.0006	—
1401	TrQA TinyLl.	LLMCHECK	22	0.688	0.0010	—
1402	TrQA TinyLl.	ROWSTAT	44	0.752	0.0001	—
1403	TrQA TinyLl.	SINK	44	0.720	0.0020	—
	TrQA TinyLl.	0D	44	0.720	0.0020	—
	TrQA TinyLl.	LEGACY MSTPROXY	22	0.729	0.0006	—
	TrQA TinyLl.	1D	88	0.659	0.0007	—
	TrQA TinyLl.	DEFL	132	0.726	0.0024	—
	TrQA TinyLl.	ANS	132	0.701	0.0021	—
	TrQA TinyLl.	PH SINK	704	0.814	0.0016	—
	TrQA TinyLl.	PH 0DTOT	704	0.792	0.0018	—
	TrQA TinyLl.	PH 0D	1408	0.775	0.0027	—
	TrQA TinyLl.	PH ENT	704	0.807	0.0003	—
	TrQA TinyLl.	LOOKBACK	704	0.823	0.0024	—
	TrQA TinyLl.	HIDDEN	2048	0.827	0.0028	—

Setting	Bank	dim	AUC	seed SD	pair acc.
TrQA TinyLl.	NONTOPO	2824	0.842	0.0031	—
TrQA TinyLl.	0D+1D	132	0.718	0.0002	—
TrQA TinyLl.	0D+SINK	88	0.724	0.0015	—
TrQA TinyLl.	HIDDEN+0D	2092	0.828	0.0027	—
TrQA TinyLl.	LEX+0D	—	0.764	0.0010	—
TrQA TinyLl.	LEX+DEFL	—	0.757	0.0032	—
TrQA TinyLl.	LEX+SINK	—	0.762	0.0021	—
TrQA TinyLl.	LOGPROB+0D	50	0.762	0.0021	—
TrQA TinyLl.	NONTOPO+0D	2868	0.842	0.0029	—
TrQA TinyLl.	NONTOPO+DEFL	2956	0.842	0.0032	—
TrQA TinyLl.	PH SINK+PH 0DTOT	1408	0.813	0.0016	—
TrQA TinyLl.	SINK+1D	132	0.719	0.0013	—
TrQA TinyLl.	SINK+ANS	176	0.725	0.0030	—
TrQA TinyLl.	SINK+DEFL	176	0.734	0.0032	—

Setting	Test (A $\rightarrow$ B)	AUC _A	AUC _B	Δ	90% CI	$\Xi_{.015}$	power
TQA Qw3B	0D $\rightarrow$ SINK	0.736	0.737	+0.001	[-0.006, +0.009]	yes	0.90
TQA Qw3B	0D $\rightarrow$ LEGACY_MSTPROXY	0.736	0.713	-0.023	[-0.034, -0.012]	no	0.42
TQA Qw3B	SINK $\rightarrow$ 0D+SINK	0.737	0.741	+0.003	[-0.003, +0.010]	yes	0.97
TQA Qw3B	SINK $\rightarrow$ SINK+DEFL	0.737	0.766	+0.029	[+0.017, +0.041]	no	0.30
TQA Qw3B	0D $\rightarrow$ 0D+1D	0.736	0.769	+0.033	[+0.022, +0.045]	no	0.38
TQA Qw3B	SINK $\rightarrow$ SINK+1D	0.737	0.766	+0.028	[+0.017, +0.039]	no	0.46
TQA Qw3B	SINK $\rightarrow$ SINK+ANS	0.737	0.785	+0.047	[+0.033, +0.062]	no	0.03
TQA Qw3B	PH_0DTOT $\rightarrow$ PH_SINK	0.842	0.851	+0.009	[+0.002, +0.016]	no	0.93
TQA Qw3B	PH_SINK $\rightarrow$ PH_SINK+PH_0DTOT	0.851	0.860	+0.009	[+0.005, +0.012]	yes	1.00
TQA Qw3B	PH_0D $\rightarrow$ PH_ENT	0.853	0.857	+0.004	[-0.006, +0.014]	yes	0.57
TQA Qw3B	0D $\rightarrow$ LOGPROB	0.736	0.630	-0.106	[-0.129, -0.083]	no	0.00
TQA Qw3B	0D $\rightarrow$ HIDDEN	0.736	0.913	+0.177	[+0.163, +0.192]	no	0.03
TQA Qw3B	0D $\rightarrow$ LOOKBACK	0.736	0.897	+0.161	[+0.146, +0.175]	no	0.02
TQA Qw3B	0D $\rightarrow$ LLMCHECK	0.736	0.693	-0.043	[-0.059, -0.026]	no	0.00
TQA Qw3B	0D $\rightarrow$ LEX	0.736	0.820	+0.084	[+0.066, +0.102]	no	0.00
TQA Qw3B	0D $\rightarrow$ ROWSTAT	0.736	0.750	+0.014	[+0.002, +0.026]	no	0.30
TQA Qw3B	0D $\rightarrow$ LEN	0.736	0.529	-0.207	[-0.229, -0.185]	no	0.00
TQA Qw3B	NONTOPO $\rightarrow$ NONTOPO+0D	0.926	0.926	+0.000	[-0.001, +0.001]	yes	1.00
TQA Qw3B	NONTOPO $\rightarrow$ NONTOPO+DEFL	0.926	0.928	+0.001	[-0.000, +0.003]	yes	1.00
TQA Qw3B	LEX $\rightarrow$ LEX+0D	0.820	0.840	+0.020	[+0.011, +0.030]	no	0.63
TQA Qw3B	LEX $\rightarrow$ LEX+SINK	0.820	0.843	+0.022	[+0.013, +0.033]	no	0.59
TQA Qw3B	LEX $\rightarrow$ LEX+DEFL	0.820	0.830	+0.010	[-0.003, +0.023]	no	0.19
TQA Qw3B	LOGPROB $\rightarrow$ LOGPROB+0D	0.630	0.776	+0.146	[+0.127, +0.165]	no	0.00
TQA Qw3B	HIDDEN $\rightarrow$ HIDDEN+0D	0.913	0.913	+0.000	[-0.001, +0.001]	yes	1.00
TQA Qw3B	NONTOPO $\rightarrow$ NONTOPO(+0D)	0.926	0.926	+0.000	[-0.001, +0.001]	yes	1.00
TQA Qw3B	NONTOPO $\rightarrow$ NONTOPO(+DEFL)	0.926	0.926	+0.000	[-0.001, +0.001]	yes	1.00
TQA Qw3B	NONTOPO $\rightarrow$ NONTOPO(+SINK)	0.926	0.926	+0.000	[-0.000, +0.001]	yes	1.00
TQA Qw3B	HIDDEN $\rightarrow$ HIDDEN(+0D)	0.913	0.914	+0.001	[-0.000, +0.002]	yes	1.00
TQA Qw3B	LOGPROB $\rightarrow$ LOGPROB(+0D)	0.630	0.757	+0.128	[+0.109, +0.147]	no	0.00
TQA Qw3B	SINK $\rightarrow$ SINK(+0D)	0.737	0.739	+0.002	[-0.001, +0.006]	yes	1.00
TQA Qw1.5B	0D $\rightarrow$ SINK	0.726	0.723	-0.003	[-0.012, +0.006]	yes	0.68
TQA Qw1.5B	0D $\rightarrow$ LEGACY_MSTPROXY	0.726	0.698	-0.028	[-0.040, -0.016]	no	0.36
TQA Qw1.5B	SINK $\rightarrow$ 0D+SINK	0.723	0.735	+0.012	[+0.005, +0.020]	no	0.91
TQA Qw1.5B	SINK $\rightarrow$ SINK+DEFL	0.723	0.757	+0.034	[+0.022, +0.045]	no	0.35
TQA Qw1.5B	0D $\rightarrow$ 0D+1D	0.726	0.718	-0.008	[-0.014, -0.002]	yes	0.99
TQA Qw1.5B	SINK $\rightarrow$ SINK+1D	0.723	0.715	-0.008	[-0.014, -0.002]	yes	0.99
TQA Qw1.5B	SINK $\rightarrow$ SINK+ANS	0.723	0.761	+0.038	[+0.024, +0.051]	no	0.11
TQA Qw1.5B	PH_0DTOT $\rightarrow$ PH_SINK	0.819	0.830	+0.011	[+0.003, +0.018]	no	0.90
TQA Qw1.5B	PH_SINK $\rightarrow$ PH_SINK+PH_0DTOT	0.830	0.841	+0.011	[+0.007, +0.014]	yes	1.00
TQA Qw1.5B	PH_0D $\rightarrow$ PH_ENT	0.838	0.833	-0.004	[-0.016, +0.007]	no	0.44
TQA Qw1.5B	0D $\rightarrow$ LOGPROB	0.726	0.638	-0.088	[-0.108, -0.067]	no	0.00
TQA Qw1.5B	0D $\rightarrow$ HIDDEN	0.726	0.916	+0.190	[+0.175, +0.204]	no	0.05
TQA Qw1.5B	0D $\rightarrow$ LOOKBACK	0.726	0.851	+0.125	[+0.109, +0.140]	no	0.00
TQA Qw1.5B	0D $\rightarrow$ LLMCHECK	0.726	0.666	-0.060	[-0.076, -0.044]	no	0.00
TQA Qw1.5B	0D $\rightarrow$ LEX	0.726	0.820	+0.094	[+0.076, +0.112]	no	0.00
TQA Qw1.5B	0D $\rightarrow$ ROWSTAT	0.726	0.738	+0.012	[-0.000, +0.025]	no	0.27
TQA Qw1.5B	0D $\rightarrow$ LEN	0.726	0.529	-0.197	[-0.218, -0.176]	no	0.00
TQA Qw1.5B	NONTOPO $\rightarrow$ NONTOPO+0D	0.917	0.917	+0.000	[-0.000, +0.001]	yes	1.00
TQA Qw1.5B	NONTOPO $\rightarrow$ NONTOPO+DEFL	0.917	0.917	+0.000	[-0.001, +0.001]	yes	1.00
TQA Qw1.5B	LEX $\rightarrow$ LEX+0D	0.820	0.835	+0.015	[+0.006, +0.024]	no	0.71
TQA Qw1.5B	LEX $\rightarrow$ LEX+SINK	0.820	0.833	+0.013	[+0.004, +0.022]	no	0.73
TQA Qw1.5B	LEX $\rightarrow$ LEX+DEFL	0.820	0.825	+0.005	[-0.006, +0.016]	no	0.45
TQA Qw1.5B	LOGPROB $\rightarrow$ LOGPROB+0D	0.638	0.740	+0.101	[+0.086, +0.116]	no	0.00
TQA Qw1.5B	HIDDEN $\rightarrow$ HIDDEN+0D	0.916	0.916	+0.001	[-0.000, +0.002]	yes	1.00
TQA Qw1.5B	NONTOPO $\rightarrow$ NONTOPO(+0D)	0.916	0.916	+0.000	[-0.001, +0.001]	yes	1.00
TQA Qw1.5B	NONTOPO $\rightarrow$ NONTOPO(+DEFL)	0.916	0.916	-0.000	[-0.001, +0.001]	yes	1.00
TQA Qw1.5B	NONTOPO $\rightarrow$ NONTOPO(+SINK)	0.916	0.916	-0.000	[-0.001, +0.000]	yes	1.00
TQA Qw1.5B	HIDDEN $\rightarrow$ HIDDEN(+0D)	0.915	0.915	+0.000	[-0.001, +0.001]	yes	1.00
TQA Qw1.5B	LOGPROB $\rightarrow$ LOGPROB(+0D)	0.638	0.737	+0.100	[+0.084, +0.115]	no	0.00
TQA Qw1.5B	SINK $\rightarrow$ SINK(+0D)	0.722	0.730	+0.007	[+0.002, +0.012]	yes	1.00
TQA Phi-3	0D $\rightarrow$ SINK	0.646	0.651	+0.005	[-0.003, +0.012]	yes	0.90
TQA Phi-3	0D $\rightarrow$ LEGACY_MSTPROXY	0.646	0.613	-0.033	[-0.046, -0.020]	no	0.20
TQA Phi-3	SINK $\rightarrow$ 0D+SINK	0.651	0.657	+0.006	[-0.000, +0.012]	yes	0.99
TQA Phi-3	SINK $\rightarrow$ SINK+DEFL	0.651	0.684	+0.033	[+0.020, +0.048]	no	0.10

	Setting	Test (A $\rightarrow$ B)	AUC _A	AUC _B	Δ	90% CI	$\Xi_{.015}$	power
1458								
1459	TQA Phi-3	0D $\rightarrow$ 0D+1D	0.646	0.643	-0.003	[-0.006, +0.001]	yes	1.00
1460	TQA Phi-3	SINK $\rightarrow$ SINK+1D	0.651	0.651	+0.000	[-0.004, +0.005]	yes	1.00
	TQA Phi-3	SINK $\rightarrow$ SINK+ANS	0.651	0.675	+0.024	[+0.007, +0.039]	no	0.00
1461	TQA Phi-3	PH_ODTOT $\rightarrow$ PH_SINK	0.845	0.871	+0.026	[+0.020, +0.031]	no	1.00
1462	TQA Phi-3	PH_SINK $\rightarrow$ PH_SINK+PH_ODTOT	0.871	0.871	-0.000	[-0.001, +0.001]	yes	1.00
	TQA Phi-3	PH_OD $\rightarrow$ PH_ENT	0.868	0.864	-0.004	[-0.012, +0.004]	yes	0.87
1463	TQA Phi-3	0D $\rightarrow$ LOGPROB	0.646	0.578	-0.068	[-0.092, -0.044]	no	0.00
	TQA Phi-3	0D $\rightarrow$ HIDDEN	0.646	0.933	+0.287	[+0.272, +0.302]	no	0.00
1464	TQA Phi-3	0D $\rightarrow$ LOOKBACK	0.646	0.892	+0.246	[+0.230, +0.262]	no	0.00
	TQA Phi-3	0D $\rightarrow$ LLMCHECK	0.646	0.653	+0.007	[-0.010, +0.024]	no	0.00
1465	TQA Phi-3	0D $\rightarrow$ LEX	0.646	0.820	+0.174	[+0.154, +0.193]	no	0.00
1466	TQA Phi-3	0D $\rightarrow$ ROWSTAT	0.646	0.661	+0.015	[+0.002, +0.029]	no	0.12
	TQA Phi-3	0D $\rightarrow$ LEN	0.646	0.522	-0.124	[-0.145, -0.102]	no	0.00
1467	TQA Phi-3	NONTOPO $\rightarrow$ NONTOPO+0D	0.936	0.936	+0.000	[-0.000, +0.001]	yes	1.00
	TQA Phi-3	NONTOPO $\rightarrow$ NONTOPO+DEFL	0.936	0.937	+0.001	[-0.000, +0.002]	yes	1.00
1468	TQA Phi-3	LEX $\rightarrow$ LEX+0D	0.820	0.821	+0.001	[-0.007, +0.009]	yes	0.85
	TQA Phi-3	LEX $\rightarrow$ LEX+SINK	0.820	0.820	-0.000	[-0.008, +0.007]	yes	0.89
1469	TQA Phi-3	LEX $\rightarrow$ LEX+DEFL	0.820	0.802	-0.018	[-0.028, -0.007]	no	0.52
1470	TQA Phi-3	LOGPROB $\rightarrow$ LOGPROB+0D	0.578	0.655	+0.077	[+0.058, +0.096]	no	0.00
	TQA Phi-3	HIDDEN $\rightarrow$ HIDDEN+0D	0.933	0.933	+0.000	[-0.001, +0.001]	yes	1.00
1471	TQA Phi-3	NONTOPO $\rightarrow$ NONTOPO(+0D)	0.936	0.936	-0.000	[-0.000, -0.000]	yes	1.00
	TQA Phi-3	NONTOPO $\rightarrow$ NONTOPO(+0D)+DEFL	0.936	0.936	-0.000	[-0.000, -0.000]	yes	1.00
1472	TQA Phi-3	NONTOPO $\rightarrow$ NONTOPO(+0D)+SINK	0.936	0.936	-0.000	[-0.000, +0.000]	yes	1.00
1473	TQA Phi-3	HIDDEN $\rightarrow$ HIDDEN(+0D)	0.932	0.932	-0.000	[-0.000, -0.000]	yes	1.00
	TQA Phi-3	LOGPROB $\rightarrow$ LOGPROB(+0D)	0.577	0.650	+0.073	[+0.055, +0.091]	no	0.00
1474	TQA Phi-3	SINK $\rightarrow$ SINK(+0D)	0.650	0.651	+0.000	[-0.003, +0.003]	yes	1.00
	TQA TinyLl.	0D $\rightarrow$ SINK	0.654	0.643	-0.010	[-0.021, +0.000]	no	0.51
1475	TQA TinyLl.	0D $\rightarrow$ LEGACY_MSTPROXY	0.654	0.627	-0.027	[-0.037, -0.016]	no	0.46
	TQA TinyLl.	SINK $\rightarrow$ 0D+SINK	0.643	0.663	+0.020	[+0.011, +0.029]	no	0.73
1476	TQA TinyLl.	SINK $\rightarrow$ SINK+DEFL	0.643	0.688	+0.045	[+0.032, +0.057]	no	0.28
1477	TQA TinyLl.	0D $\rightarrow$ 0D+1D	0.654	0.660	+0.007	[-0.000, +0.013]	yes	0.95
	TQA TinyLl.	SINK $\rightarrow$ SINK+1D	0.643	0.651	+0.008	[+0.001, +0.015]	no	0.93
1478	TQA TinyLl.	SINK $\rightarrow$ SINK+ANS	0.643	0.685	+0.041	[+0.025, +0.058]	no	0.00
	TQA TinyLl.	PH_ODTOT $\rightarrow$ PH_SINK	0.804	0.808	+0.004	[-0.002, +0.011]	yes	0.97
1479	TQA TinyLl.	PH_SINK $\rightarrow$ PH_SINK+PH_ODTOT	0.808	0.810	+0.003	[+0.000, +0.005]	yes	1.00
	TQA TinyLl.	PH_OD $\rightarrow$ PH_ENT	0.806	0.817	+0.011	[+0.000, +0.022]	no	0.50
1480	TQA TinyLl.	0D $\rightarrow$ LOGPROB	0.654	0.637	-0.017	[-0.039, +0.004]	no	0.00
	TQA TinyLl.	0D $\rightarrow$ HIDDEN	0.654	0.883	+0.229	[+0.214, +0.244]	no	0.00
1482	TQA TinyLl.	0D $\rightarrow$ LOOKBACK	0.654	0.850	+0.197	[+0.182, +0.212]	no	0.00
	TQA TinyLl.	0D $\rightarrow$ LLMCHECK	0.654	0.614	-0.039	[-0.055, -0.024]	no	0.00
1483	TQA TinyLl.	0D $\rightarrow$ LEX	0.654	0.820	+0.166	[+0.149, +0.184]	no	0.00
	TQA TinyLl.	0D $\rightarrow$ ROWSTAT	0.654	0.661	+0.007	[-0.004, +0.019]	no	0.37
1484	TQA TinyLl.	0D $\rightarrow$ LEN	0.654	0.523	-0.131	[-0.152, -0.110]	no	0.00
	TQA TinyLl.	NONTOPO $\rightarrow$ NONTOPO+0D	0.885	0.885	+0.000	[-0.000, +0.000]	yes	1.00
1485	TQA TinyLl.	NONTOPO $\rightarrow$ NONTOPO+DEFL	0.885	0.884	-0.001	[-0.002, +0.000]	yes	1.00
1486	TQA TinyLl.	LEX $\rightarrow$ LEX+0D	0.820	0.830	+0.010	[+0.003, +0.016]	no	0.97
	TQA TinyLl.	LEX $\rightarrow$ LEX+SINK	0.820	0.831	+0.011	[+0.005, +0.017]	no	0.98
1487	TQA TinyLl.	LEX $\rightarrow$ LEX+DEFL	0.820	0.826	+0.005	[-0.003, +0.014]	yes	0.75
	TQA TinyLl.	LOGPROB $\rightarrow$ LOGPROB+0D	0.637	0.683	+0.046	[+0.033, +0.060]	no	0.13
1488	TQA TinyLl.	HIDDEN $\rightarrow$ HIDDEN+0D	0.883	0.883	-0.000	[-0.001, +0.000]	yes	1.00
	TQA TinyLl.	NONTOPO $\rightarrow$ NONTOPO(+0D)	0.884	0.884	-0.000	[-0.000, -0.000]	yes	1.00
1489	TQA TinyLl.	NONTOPO $\rightarrow$ NONTOPO(+0D)+DEFL	0.884	0.884	-0.000	[-0.001, +0.000]	yes	1.00
	TQA TinyLl.	NONTOPO $\rightarrow$ NONTOPO(+0D)+SINK	0.884	0.884	-0.000	[-0.000, -0.000]	yes	1.00
1490	TQA TinyLl.	HIDDEN $\rightarrow$ HIDDEN(+0D)	0.883	0.882	-0.000	[-0.000, +0.000]	yes	1.00
1491	TQA TinyLl.	LOGPROB $\rightarrow$ LOGPROB(+0D)	0.636	0.678	+0.043	[+0.030, +0.055]	no	0.25
1492	TQA TinyLl.	SINK $\rightarrow$ SINK(+0D)	0.643	0.657	+0.014	[+0.007, +0.021]	no	0.96
	TQA SmolLM	0D $\rightarrow$ SINK	0.583	0.611	+0.028	[+0.009, +0.048]	no	0.00
1493	TQA SmolLM	0D $\rightarrow$ LEGACY_MSTPROXY	0.583	0.585	+0.002	[-0.003, +0.006]	yes	1.00
	TQA SmolLM	SINK $\rightarrow$ 0D+SINK	0.611	0.641	+0.030	[+0.021, +0.039]	no	0.66
1494	TQA SmolLM	SINK $\rightarrow$ SINK+DEFL	0.611	0.652	+0.041	[+0.030, +0.053]	no	0.34
	TQA SmolLM	0D $\rightarrow$ 0D+1D	0.583	0.588	+0.005	[-0.004, +0.014]	yes	0.68
1495	TQA SmolLM	SINK $\rightarrow$ SINK+1D	0.611	0.629	+0.018	[+0.008, +0.028]	no	0.62
	TQA SmolLM	SINK $\rightarrow$ SINK+ANS	0.611	0.646	+0.036	[+0.018, +0.053]	no	0.00
1496	TQA SmolLM	PH_ODTOT $\rightarrow$ PH_SINK	0.819	0.720	-0.099	[-0.113, -0.086]	no	0.15
1497	TQA SmolLM	PH_SINK $\rightarrow$ PH_SINK+PH_ODTOT	0.720	0.826	+0.106	[+0.094, +0.118]	no	0.29
	TQA SmolLM	PH_OD $\rightarrow$ PH_ENT	0.800	0.834	+0.034	[+0.024, +0.044]	no	0.60
1498	TQA SmolLM	0D $\rightarrow$ LOGPROB	0.583	0.588	+0.005	[-0.017, +0.027]	no	0.00
	TQA SmolLM	0D $\rightarrow$ HIDDEN	0.583	0.894	+0.311	[+0.296, +0.326]	no	0.01
1499	TQA SmolLM	0D $\rightarrow$ LOOKBACK	0.583	0.863	+0.280	[+0.264, +0.295]	no	0.00
	TQA SmolLM	0D $\rightarrow$ LLMCHECK	0.583	0.604	+0.021	[+0.006, +0.037]	no	0.00
1500	TQA SmolLM	0D $\rightarrow$ LEX	0.583	0.820	+0.237	[+0.218, +0.256]	no	0.00
	TQA SmolLM	0D $\rightarrow$ ROWSTAT	0.583	0.673	+0.090	[+0.079, +0.102]	no	0.37
1501	TQA SmolLM	0D $\rightarrow$ LEN	0.583	0.524	-0.059	[-0.074, -0.043]	no	0.00
1502	TQA SmolLM	NONTOPO $\rightarrow$ NONTOPO+0D	0.899	0.899	-0.000	[-0.001, +0.001]	yes	1.00
	TQA SmolLM	NONTOPO $\rightarrow$ NONTOPO+DEFL	0.899	0.899	-0.001	[-0.002, +0.000]	yes	1.00
1503	TQA SmolLM	LEX $\rightarrow$ LEX+0D	0.820	0.827	+0.007	[+0.001, +0.012]	yes	1.00
	TQA SmolLM	LEX $\rightarrow$ LEX+SINK	0.820	0.819	-0.001	[-0.007, +0.004]	yes	1.00
1504	TQA SmolLM	LEX $\rightarrow$ LEX+DEFL	0.820	0.822	+0.002	[-0.005, +0.009]	yes	0.91
1505	TQA SmolLM	LOGPROB $\rightarrow$ LOGPROB+0D	0.588	0.613	+0.025	[+0.015, +0.035]	no	0.59
	TQA SmolLM	HIDDEN $\rightarrow$ HIDDEN+0D	0.894	0.894	+0.000	[-0.001, +0.001]	yes	1.00
1506	TQA SmolLM	NONTOPO $\rightarrow$ NONTOPO(+0D)	0.899	0.899	-0.000	[-0.000, -0.000]	yes	1.00
	TQA SmolLM	NONTOPO $\rightarrow$ NONTOPO(+0D)+DEFL	0.899	0.899	-0.000	[-0.000, -0.000]	yes	1.00
1507	TQA SmolLM	NONTOPO $\rightarrow$ NONTOPO(+0D)+SINK	0.899	0.899	-0.000	[-0.000, +0.000]	yes	1.00
1508	TQA SmolLM	HIDDEN $\rightarrow$ HIDDEN(+0D)	0.893	0.893	-0.000	[-0.001, -0.000]	yes	1.00
	TQA SmolLM	LOGPROB $\rightarrow$ LOGPROB(+0D)	0.588	0.607	+0.019	[+0.010, +0.029]	no	0.67
1509	TQA SmolLM	SINK $\rightarrow$ SINK(+0D)	0.610	0.633	+0.022	[+0.014, +0.031]	no	0.81
	HE Qw3B	0D $\rightarrow$ SINK	0.811	0.829	+0.018	[+0.008, +0.028]	no	0.64
1510	HE Qw3B	0D $\rightarrow$ LEGACY_MSTPROXY	0.811	0.731	-0.080	[-0.091, -0.069]	no	0.46
	HE Qw3B	SINK $\rightarrow$ 0D+SINK	0.829	0.857	+0.028	[+0.022, +0.035]	no	0.97
1511	HE Qw3B	SINK $\rightarrow$ SINK+DEFL	0.829	0.887	+0.058	[+0.049, +0.067]	no	0.73

Setting	Test (A $\rightarrow$ B)	AUC _A	AUC _B	Δ	90% CI	$\Xi_{.015}$	power
HE Qw3B	0D $\rightarrow$ 0D+1D	0.811	0.826	+0.015	[+0.009, +0.023]	no	0.94
HE Qw3B	SINK $\rightarrow$ SINK+1D	0.829	0.843	+0.014	[+0.007, +0.021]	no	0.97
HE Qw3B	SINK $\rightarrow$ SINK+ANS	0.829	0.978	+0.150	[+0.139, +0.161]	no	0.44
HE Qw3B	PH_0DTOT $\rightarrow$ PH_SINK	0.958	0.958	+0.000	[-0.005, +0.006]	yes	1.00
HE Qw3B	PH_SINK $\rightarrow$ PH_SINK+PH_0DTOT	0.958	0.962	+0.004	[+0.003, +0.005]	yes	1.00
HE Qw3B	PH_0D $\rightarrow$ PH_ENT	0.971	0.964	-0.006	[-0.012, -0.000]	yes	0.99
HE Qw3B	0D $\rightarrow$ LOGPROB	0.811	0.993	+0.182	[+0.170, +0.194]	no	0.33
HE Qw3B	0D $\rightarrow$ HIDDEN	0.811	1.000	+0.189	[+0.177, +0.201]	no	0.32
HE Qw3B	0D $\rightarrow$ LOOKBACK	0.811	0.998	+0.187	[+0.175, +0.199]	no	0.32
HE Qw3B	0D $\rightarrow$ LLMCHECK	0.811	0.745	-0.066	[-0.078, -0.054]	no	0.27
HE Qw3B	0D $\rightarrow$ LEX	0.811	0.981	+0.170	[+0.159, +0.182]	no	0.37
HE Qw3B	0D $\rightarrow$ ROWSTAT	0.811	0.829	+0.018	[+0.007, +0.030]	no	0.35
HE Qw3B	0D $\rightarrow$ LEN	0.811	0.962	+0.151	[+0.140, +0.163]	no	0.36
HE Qw3B	NONTOPO $\rightarrow$ NONTOPO+0D	1.000	1.000	+0.000	[-0.000, +0.000]	yes	1.00
HE Qw3B	NONTOPO $\rightarrow$ NONTOPO+DEFL	1.000	1.000	+0.000	[-0.000, +0.000]	yes	1.00
HE Qw3B	LEX $\rightarrow$ LEX+0D	0.981	0.979	-0.002	[-0.005, +0.000]	yes	1.00
HE Qw3B	LEX $\rightarrow$ LEX+SINK	0.981	0.980	-0.001	[-0.003, +0.001]	yes	1.00
HE Qw3B	LEX $\rightarrow$ LEX+DEFL	0.981	0.971	-0.010	[-0.014, -0.007]	yes	1.00
HE Qw3B	LOGPROB $\rightarrow$ LOGPROB+0D	0.993	0.993	+0.000	[-0.001, +0.001]	yes	1.00
HE Qw3B	HIDDEN $\rightarrow$ HIDDEN+0D	1.000	1.000	-0.000	[-0.000, +0.000]	yes	1.00
HE Qw3B	NONTOPO $\rightarrow$ NONTOPO(+0D)	1.000	1.000	-0.000	[-0.000, +0.000]	yes	1.00
HE Qw3B	NONTOPO $\rightarrow$ NONTOPO(+DEFL)	1.000	1.000	-0.000	[-0.000, +0.000]	yes	1.00
HE Qw3B	NONTOPO $\rightarrow$ NONTOPO(+SINK)	1.000	1.000	-0.000	[-0.000, +0.000]	yes	1.00
HE Qw3B	HIDDEN $\rightarrow$ HIDDEN(+0D)	1.000	1.000	+0.000	[-0.000, +0.000]	yes	1.00
HE Qw3B	LOGPROB $\rightarrow$ LOGPROB(+0D)	0.993	0.993	+0.001	[+0.000, +0.001]	yes	1.00
HE Qw3B	SINK $\rightarrow$ SINK(+0D)	0.828	0.839	+0.011	[+0.007, +0.015]	yes	1.00
HE Qw1.5B	0D $\rightarrow$ SINK	0.789	0.820	+0.031	[+0.017, +0.045]	no	0.08
HE Qw1.5B	0D $\rightarrow$ LEGACY_MSTPROXY	0.789	0.719	-0.070	[-0.086, -0.053]	no	0.00
HE Qw1.5B	SINK $\rightarrow$ 0D+SINK	0.820	0.832	+0.012	[+0.005, +0.019]	no	0.91
HE Qw1.5B	SINK $\rightarrow$ SINK+DEFL	0.820	0.872	+0.052	[+0.038, +0.065]	no	0.17
HE Qw1.5B	0D $\rightarrow$ 0D+1D	0.789	0.815	+0.026	[+0.013, +0.040]	no	0.17
HE Qw1.5B	SINK $\rightarrow$ SINK+1D	0.820	0.826	+0.006	[-0.005, +0.017]	no	0.45
HE Qw1.5B	SINK $\rightarrow$ SINK+ANS	0.820	0.975	+0.155	[+0.138, +0.172]	no	0.00
HE Qw1.5B	PH_0DTOT $\rightarrow$ PH_SINK	0.921	0.922	+0.002	[-0.009, +0.013]	yes	0.46
HE Qw1.5B	PH_SINK $\rightarrow$ PH_SINK+PH_0DTOT	0.922	0.931	+0.008	[+0.006, +0.011]	yes	1.00
HE Qw1.5B	PH_0D $\rightarrow$ PH_ENT	0.949	0.933	-0.016	[-0.028, -0.004]	no	0.33
HE Qw1.5B	0D $\rightarrow$ LOGPROB	0.789	0.995	+0.205	[+0.188, +0.222]	no	0.00
HE Qw1.5B	0D $\rightarrow$ HIDDEN	0.789	1.000	+0.210	[+0.193, +0.227]	no	0.00
HE Qw1.5B	0D $\rightarrow$ LOOKBACK	0.789	0.999	+0.210	[+0.193, +0.226]	no	0.00
HE Qw1.5B	0D $\rightarrow$ LLMCHECK	0.789	0.791	+0.002	[-0.017, +0.021]	no	0.00
HE Qw1.5B	0D $\rightarrow$ LEX	0.789	0.979	+0.190	[+0.172, +0.207]	no	0.00
HE Qw1.5B	0D $\rightarrow$ ROWSTAT	0.789	0.788	-0.001	[-0.018, +0.016]	no	0.00
HE Qw1.5B	0D $\rightarrow$ LEN	0.789	0.967	+0.178	[+0.160, +0.196]	no	0.00
HE Qw1.5B	NONTOPO $\rightarrow$ NONTOPO+0D	1.000	1.000	+0.000	[+0.000, +0.000]	yes	1.00
HE Qw1.5B	NONTOPO $\rightarrow$ NONTOPO+DEFL	1.000	1.000	+0.000	[-0.000, +0.000]	yes	1.00
HE Qw1.5B	LEX $\rightarrow$ LEX+0D	0.979	0.971	-0.008	[-0.014, -0.002]	yes	0.99
HE Qw1.5B	LEX $\rightarrow$ LEX+SINK	0.979	0.968	-0.011	[-0.017, -0.005]	no	0.98
HE Qw1.5B	LEX $\rightarrow$ LEX+DEFL	0.979	0.956	-0.022	[-0.030, -0.016]	no	0.94
HE Qw1.5B	LOGPROB $\rightarrow$ LOGPROB+0D	0.995	0.994	-0.001	[-0.003, +0.001]	yes	1.00
HE Qw1.5B	HIDDEN $\rightarrow$ HIDDEN+0D	1.000	1.000	+0.000	[-0.000, +0.000]	yes	1.00
HE Qw1.5B	NONTOPO $\rightarrow$ NONTOPO(+0D)	1.000	1.000	+0.000	[-0.000, +0.000]	yes	1.00
HE Qw1.5B	NONTOPO $\rightarrow$ NONTOPO(+DEFL)	1.000	1.000	-0.000	[-0.000, +0.000]	yes	1.00
HE Qw1.5B	NONTOPO $\rightarrow$ NONTOPO(+SINK)	1.000	1.000	+0.000	[-0.000, +0.000]	yes	1.00
HE Qw1.5B	HIDDEN $\rightarrow$ HIDDEN(+0D)	1.000	1.000	-0.000	[-0.000, +0.000]	yes	1.00
HE Qw1.5B	LOGPROB $\rightarrow$ LOGPROB(+0D)	0.994	0.994	-0.000	[-0.001, +0.001]	yes	1.00
HE Qw1.5B	SINK $\rightarrow$ SINK(+0D)	0.819	0.823	+0.004	[-0.000, +0.009]	yes	1.00
TrQA Qw3B	0D $\rightarrow$ SINK	0.732	0.717	-0.015	[-0.026, -0.004]	no	0.48
TrQA Qw3B	0D $\rightarrow$ LEGACY_MSTPROXY	0.732	0.691	-0.041	[-0.057, -0.025]	no	0.00
TrQA Qw3B	SINK $\rightarrow$ 0D+SINK	0.717	0.738	+0.021	[+0.012, +0.030]	no	0.76
TrQA Qw3B	SINK $\rightarrow$ SINK+DEFL	0.717	0.719	+0.002	[-0.012, +0.015]	no	0.13
TrQA Qw3B	0D $\rightarrow$ 0D+1D	0.732	0.729	-0.003	[-0.012, +0.006]	yes	0.74
TrQA Qw3B	SINK $\rightarrow$ SINK+1D	0.717	0.718	+0.001	[-0.009, +0.010]	yes	0.65
TrQA Qw3B	SINK $\rightarrow$ SINK+ANS	0.717	0.752	+0.035	[+0.019, +0.051]	no	0.00
TrQA Qw3B	PH_0DTOT $\rightarrow$ PH_SINK	0.807	0.820	+0.012	[+0.004, +0.021]	no	0.82
TrQA Qw3B	PH_SINK $\rightarrow$ PH_SINK+PH_0DTOT	0.820	0.824	+0.004	[+0.002, +0.006]	yes	1.00
TrQA Qw3B	PH_0D $\rightarrow$ PH_ENT	0.807	0.810	+0.003	[-0.011, +0.017]	no	0.08
TrQA Qw3B	0D $\rightarrow$ LOGPROB	0.732	0.839	+0.107	[+0.088, +0.125]	no	0.00
TrQA Qw3B	0D $\rightarrow$ HIDDEN	0.732	0.841	+0.109	[+0.090, +0.129]	no	0.00
TrQA Qw3B	0D $\rightarrow$ LOOKBACK	0.732	0.861	+0.129	[+0.111, +0.147]	no	0.00
TrQA Qw3B	0D $\rightarrow$ LLMCHECK	0.732	0.688	-0.044	[-0.065, -0.023]	no	0.00
TrQA Qw3B	0D $\rightarrow$ LEX	0.732	0.596	-0.137	[-0.163, -0.110]	no	0.00
TrQA Qw3B	0D $\rightarrow$ ROWSTAT	0.732	0.724	-0.009	[-0.025, +0.008]	no	0.00
TrQA Qw3B	0D $\rightarrow$ LEN	0.732	0.569	-0.163	[-0.189, -0.137]	no	0.00
TrQA Qw3B	NONTOPO $\rightarrow$ NONTOPO+0D	0.870	0.870	+0.001	[-0.000, +0.002]	yes	1.00
TrQA Qw3B	NONTOPO $\rightarrow$ NONTOPO+DEFL	0.870	0.870	+0.000	[-0.001, +0.002]	yes	1.00
TrQA Qw3B	LEX $\rightarrow$ LEX+0D	0.596	0.733	+0.137	[+0.117, +0.158]	no	0.00
TrQA Qw3B	LEX $\rightarrow$ LEX+SINK	0.596	0.724	+0.128	[+0.108, +0.149]	no	0.00
TrQA Qw3B	LEX $\rightarrow$ LEX+DEFL	0.596	0.690	+0.095	[+0.073, +0.116]	no	0.00
TrQA Qw3B	LOGPROB $\rightarrow$ LOGPROB+0D	0.839	0.843	+0.004	[-0.002, +0.011]	yes	0.95
TrQA Qw3B	HIDDEN $\rightarrow$ HIDDEN+0D	0.841	0.843	+0.002	[+0.000, +0.004]	yes	1.00
TrQA Qw3B	NONTOPO $\rightarrow$ NONTOPO(+0D)	0.869	0.871	+0.002	[-0.000, +0.004]	yes	1.00
TrQA Qw3B	NONTOPO $\rightarrow$ NONTOPO(+DEFL)	0.869	0.870	+0.001	[-0.001, +0.002]	yes	1.00
TrQA Qw3B	NONTOPO $\rightarrow$ NONTOPO(+SINK)	0.869	0.871	+0.001	[-0.001, +0.004]	yes	1.00
TrQA Qw3B	HIDDEN $\rightarrow$ HIDDEN(+0D)	0.841	0.849	+0.008	[+0.003, +0.012]	yes	1.00
TrQA Qw3B	LOGPROB $\rightarrow$ LOGPROB(+0D)	0.838	0.844	+0.006	[+0.002, +0.010]	yes	1.00
TrQA Qw3B	SINK $\rightarrow$ SINK(+0D)	0.716	0.733	+0.017	[+0.010, +0.025]	no	0.89
TrQA Qw1.5B	0D $\rightarrow$ SINK	0.716	0.707	-0.009	[-0.020, +0.002]	no	0.43
TrQA Qw1.5B	0D $\rightarrow$ LEGACY_MSTPROXY	0.716	0.685	-0.031	[-0.046, -0.015]	no	0.00
TrQA Qw1.5B	SINK $\rightarrow$ 0D+SINK	0.707	0.717	+0.010	[+0.001, +0.020]	no	0.68
TrQA Qw1.5B	SINK $\rightarrow$ SINK+DEFL	0.707	0.719	+0.012	[-0.003, +0.027]	no	0.00

Setting	Test (A $\rightarrow$ B)	AUC _A	AUC _B	Δ	90% CI	$\Xi_{.015}$	power
TrQA Qw1.5B	0D $\rightarrow$ 0D+1D	0.716	0.714	-0.002	[-0.011, +0.008]	yes	0.65
TrQA Qw1.5B	SINK $\rightarrow$ SINK+1D	0.707	0.709	+0.002	[-0.008, +0.013]	yes	0.49
TrQA Qw1.5B	SINK $\rightarrow$ SINK+ANS	0.707	0.726	+0.019	[+0.001, +0.036]	no	0.00
TrQA Qw1.5B	PH_0DTOT $\rightarrow$ PH_SINK	0.759	0.767	+0.008	[-0.003, +0.018]	no	0.53
TrQA Qw1.5B	PH_SINK $\rightarrow$ PH_SINK+PH_0DTOT	0.767	0.772	+0.004	[+0.001, +0.008]	yes	1.00
TrQA Qw1.5B	PH_0D $\rightarrow$ PH_ENT	0.768	0.768	-0.000	[-0.017, +0.017]	no	0.00
TrQA Qw1.5B	0D $\rightarrow$ LOGPROB	0.716	0.823	+0.107	[+0.085, +0.129]	no	0.00
TrQA Qw1.5B	0D $\rightarrow$ HIDDEN	0.716	0.855	+0.140	[+0.119, +0.160]	no	0.00
TrQA Qw1.5B	0D $\rightarrow$ LOOKBACK	0.716	0.824	+0.108	[+0.088, +0.129]	no	0.00
TrQA Qw1.5B	0D $\rightarrow$ LLMCHECK	0.716	0.676	-0.039	[-0.062, -0.016]	no	0.00
TrQA Qw1.5B	0D $\rightarrow$ LEX	0.716	0.609	-0.106	[-0.136, -0.078]	no	0.00
TrQA Qw1.5B	0D $\rightarrow$ ROWSTAT	0.716	0.720	+0.005	[-0.011, +0.020]	no	0.00
TrQA Qw1.5B	0D $\rightarrow$ LEN	0.716	0.588	-0.128	[-0.154, -0.102]	no	0.00
TrQA Qw1.5B	NONTOPO $\rightarrow$ NONTOPO+0D	0.873	0.873	+0.000	[-0.001, +0.001]	yes	1.00
TrQA Qw1.5B	NONTOPO $\rightarrow$ NONTOPO+DEFL	0.873	0.872	-0.001	[-0.002, +0.001]	yes	1.00
TrQA Qw1.5B	LEX $\rightarrow$ LEX+0D	0.609	0.721	+0.112	[+0.091, +0.134]	no	0.00
TrQA Qw1.5B	LEX $\rightarrow$ LEX+SINK	0.609	0.715	+0.106	[+0.085, +0.127]	no	0.00
TrQA Qw1.5B	LEX $\rightarrow$ LEX+DEFL	0.609	0.672	+0.063	[+0.040, +0.085]	no	0.00
TrQA Qw1.5B	LOGPROB $\rightarrow$ LOGPROB+0D	0.823	0.828	+0.006	[-0.001, +0.013]	yes	0.92
TrQA Qw1.5B	HIDDEN $\rightarrow$ HIDDEN+0D	0.855	0.855	+0.000	[-0.001, +0.002]	yes	1.00
TrQA Qw1.5B	NONTOPO $\rightarrow$ NONTOPO(+0D)	0.872	0.873	+0.000	[-0.001, +0.002]	yes	1.00
TrQA Qw1.5B	NONTOPO $\rightarrow$ NONTOPO(+0D)+DEFL	0.872	0.873	+0.000	[-0.001, +0.002]	yes	1.00
TrQA Qw1.5B	NONTOPO $\rightarrow$ NONTOPO(+0D)+SINK	0.872	0.873	+0.001	[-0.001, +0.002]	yes	1.00
TrQA Qw1.5B	HIDDEN $\rightarrow$ HIDDEN(+0D)	0.855	0.857	+0.002	[-0.000, +0.004]	yes	1.00
TrQA Qw1.5B	LOGPROB $\rightarrow$ LOGPROB(+0D)	0.822	0.828	+0.006	[+0.001, +0.010]	yes	1.00
TrQA Qw1.5B	SINK $\rightarrow$ SINK(+0D)	0.706	0.717	+0.011	[+0.003, +0.018]	no	0.93
TrQA Phi-3	0D $\rightarrow$ SINK	0.710	0.707	-0.004	[-0.010, +0.002]	yes	0.98
TrQA Phi-3	0D $\rightarrow$ LEGACY_MSTPROXY	0.710	0.669	-0.041	[-0.057, -0.025]	no	0.00
TrQA Phi-3	SINK $\rightarrow$ 0D+SINK	0.707	0.713	+0.006	[+0.000, +0.012]	yes	0.99
TrQA Phi-3	SINK $\rightarrow$ SINK+DEFL	0.707	0.736	+0.029	[+0.013, +0.045]	no	0.00
TrQA Phi-3	0D $\rightarrow$ 0D+1D	0.710	0.705	-0.005	[-0.011, +0.000]	yes	0.99
TrQA Phi-3	SINK $\rightarrow$ SINK+1D	0.707	0.699	-0.008	[-0.013, -0.002]	yes	1.00
TrQA Phi-3	SINK $\rightarrow$ SINK+ANS	0.707	0.736	+0.029	[+0.012, +0.046]	no	0.00
TrQA Phi-3	PH_0DTOT $\rightarrow$ PH_SINK	0.810	0.826	+0.016	[+0.009, +0.022]	no	0.96
TrQA Phi-3	PH_SINK $\rightarrow$ PH_SINK+PH_0DTOT	0.826	0.828	+0.002	[-0.001, +0.004]	yes	1.00
TrQA Phi-3	PH_0D $\rightarrow$ PH_ENT	0.807	0.817	+0.010	[-0.002, +0.022]	no	0.31
TrQA Phi-3	0D $\rightarrow$ LOGPROB	0.710	0.877	+0.167	[+0.148, +0.186]	no	0.00
TrQA Phi-3	0D $\rightarrow$ HIDDEN	0.710	0.867	+0.157	[+0.137, +0.176]	no	0.00
TrQA Phi-3	0D $\rightarrow$ LOOKBACK	0.710	0.840	+0.129	[+0.109, +0.150]	no	0.00
TrQA Phi-3	0D $\rightarrow$ LLMCHECK	0.710	0.676	-0.034	[-0.055, -0.014]	no	0.00
TrQA Phi-3	0D $\rightarrow$ LEX	0.710	0.635	-0.076	[-0.102, -0.050]	no	0.00
TrQA Phi-3	0D $\rightarrow$ ROWSTAT	0.710	0.728	+0.018	[+0.002, +0.034]	no	0.00
TrQA Phi-3	0D $\rightarrow$ LEN	0.710	0.581	-0.130	[-0.156, -0.104]	no	0.00
TrQA Phi-3	NONTOPO $\rightarrow$ NONTOPO+0D	0.883	0.883	-0.000	[-0.001, +0.000]	yes	1.00
TrQA Phi-3	NONTOPO $\rightarrow$ NONTOPO+DEFL	0.883	0.885	+0.002	[+0.000, +0.004]	yes	1.00
TrQA Phi-3	LEX $\rightarrow$ LEX+0D	0.635	0.721	+0.086	[+0.069, +0.104]	no	0.00
TrQA Phi-3	LEX $\rightarrow$ LEX+SINK	0.635	0.718	+0.083	[+0.066, +0.101]	no	0.00
TrQA Phi-3	LEX $\rightarrow$ LEX+DEFL	0.635	0.712	+0.077	[+0.057, +0.098]	no	0.00
TrQA Phi-3	LOGPROB $\rightarrow$ LOGPROB+0D	0.877	0.878	+0.000	[-0.002, +0.002]	yes	1.00
TrQA Phi-3	HIDDEN $\rightarrow$ HIDDEN+0D	0.867	0.867	-0.000	[-0.001, +0.000]	yes	1.00
TrQA Phi-3	NONTOPO $\rightarrow$ NONTOPO(+0D)	0.883	0.884	+0.000	[-0.001, +0.002]	yes	1.00
TrQA Phi-3	NONTOPO $\rightarrow$ NONTOPO(+0D)+DEFL	0.883	0.884	+0.001	[-0.001, +0.003]	yes	1.00
TrQA Phi-3	NONTOPO $\rightarrow$ NONTOPO(+0D)+SINK	0.883	0.883	+0.000	[-0.001, +0.001]	yes	1.00
TrQA Phi-3	HIDDEN $\rightarrow$ HIDDEN(+0D)	0.867	0.869	+0.002	[-0.000, +0.004]	yes	1.00
TrQA Phi-3	LOGPROB $\rightarrow$ LOGPROB(+0D)	0.877	0.878	+0.001	[-0.001, +0.003]	yes	1.00
TrQA Phi-3	SINK $\rightarrow$ SINK(+0D)	0.706	0.710	+0.004	[-0.001, +0.009]	yes	1.00
TrQA TinyLL	0D $\rightarrow$ SINK	0.720	0.720	+0.001	[-0.010, +0.012]	yes	0.40
TrQA TinyLL	0D $\rightarrow$ LEGACY_MSTPROXY	0.720	0.729	+0.010	[-0.000, +0.020]	no	0.55
TrQA TinyLL	SINK $\rightarrow$ 0D+SINK	0.720	0.724	+0.004	[-0.003, +0.011]	yes	0.93
TrQA TinyLL	SINK $\rightarrow$ SINK+DEFL	0.720	0.734	+0.014	[-0.000, +0.028]	no	0.10
TrQA TinyLL	0D $\rightarrow$ 0D+1D	0.720	0.718	-0.002	[-0.011, +0.008]	yes	0.60
TrQA TinyLL	SINK $\rightarrow$ SINK+1D	0.720	0.719	-0.001	[-0.010, +0.008]	yes	0.73
TrQA TinyLL	SINK $\rightarrow$ SINK+ANS	0.720	0.725	+0.005	[-0.011, +0.021]	no	0.00
TrQA TinyLL	PH_0DTOT $\rightarrow$ PH_SINK	0.792	0.814	+0.022	[+0.015, +0.029]	no	0.94
TrQA TinyLL	PH_SINK $\rightarrow$ PH_SINK+PH_0DTOT	0.814	0.813	-0.001	[-0.002, +0.000]	yes	1.00
TrQA TinyLL	PH_0D $\rightarrow$ PH_ENT	0.775	0.807	+0.032	[+0.017, +0.047]	no	0.01
TrQA TinyLL	0D $\rightarrow$ LOGPROB	0.720	0.724	+0.004	[-0.019, +0.027]	no	0.00
TrQA TinyLL	0D $\rightarrow$ HIDDEN	0.720	0.827	+0.108	[+0.088, +0.128]	no	0.00
TrQA TinyLL	0D $\rightarrow$ LOOKBACK	0.720	0.823	+0.104	[+0.087, +0.121]	no	0.00
TrQA TinyLL	0D $\rightarrow$ LLMCHECK	0.720	0.688	-0.032	[-0.050, -0.013]	no	0.00
TrQA TinyLL	0D $\rightarrow$ LEX	0.720	0.740	+0.021	[-0.003, +0.044]	no	0.00
TrQA TinyLL	0D $\rightarrow$ ROWSTAT	0.720	0.752	+0.032	[+0.020, +0.045]	no	0.26
TrQA TinyLL	0D $\rightarrow$ LEN	0.720	0.631	-0.089	[-0.109, -0.068]	no	0.00
TrQA TinyLL	NONTOPO $\rightarrow$ NONTOPO+0D	0.842	0.842	-0.000	[-0.001, +0.001]	yes	1.00
TrQA TinyLL	NONTOPO $\rightarrow$ NONTOPO+DEFL	0.842	0.842	+0.000	[-0.003, +0.003]	yes	1.00
TrQA TinyLL	LEX $\rightarrow$ LEX+0D	0.740	0.764	+0.024	[+0.012, +0.036]	no	0.33
TrQA TinyLL	LEX $\rightarrow$ LEX+SINK	0.740	0.762	+0.021	[+0.010, +0.032]	no	0.45
TrQA TinyLL	LEX $\rightarrow$ LEX+DEFL	0.740	0.757	+0.016	[+0.002, +0.031]	no	0.04
TrQA TinyLL	LOGPROB $\rightarrow$ LOGPROB+0D	0.724	0.762	+0.038	[+0.025, +0.052]	no	0.17
TrQA TinyLL	HIDDEN $\rightarrow$ HIDDEN+0D	0.827	0.828	+0.001	[-0.001, +0.002]	yes	1.00
TrQA TinyLL	NONTOPO $\rightarrow$ NONTOPO(+0D)	0.842	0.842	+0.000	[-0.001, +0.002]	yes	1.00
TrQA TinyLL	NONTOPO $\rightarrow$ NONTOPO(+0D)+DEFL	0.842	0.843	+0.001	[-0.001, +0.004]	yes	1.00
TrQA TinyLL	NONTOPO $\rightarrow$ NONTOPO(+0D)+SINK	0.842	0.842	+0.001	[-0.001, +0.002]	yes	1.00
TrQA TinyLL	HIDDEN $\rightarrow$ HIDDEN(+0D)	0.827	0.828	+0.001	[-0.001, +0.004]	yes	1.00
TrQA TinyLL	LOGPROB $\rightarrow$ LOGPROB(+0D)	0.723	0.755	+0.031	[+0.020, +0.043]	no	0.40
TrQA TinyLL	SINK $\rightarrow$ SINK(+0D)	0.720	0.725	+0.005	[+0.001, +0.009]	yes	1.00

F INTERVENING ON THE SINK

Proposition 1 predicts that TOHA should track the response sink score more closely as the sink strengthens. We test this by intervening on the sink directly. For Qwen2.5-1.5B and TinyLlama on TruthfulQA, we add a constant $b \in \{-2, 0, 2, 4\}$ to the first token’s attention logit in every head and recompute all per-head quantities (Table 13). The intervention leaves the identity intact (no violations in any run). It moves the structure the proposition says matters. Suppressing the sink ($b=-2$) lowers the share of prompt-coned head graphs from 58% to 37% (Qwen) and from 74% to 63% (TinyLlama). It lowers the share where the sink is every response token’s top prompt token from 45% to 21% and from 63% to 21%, and the median rank correlation between d and the response sink score from 0.893 to 0.605 and from 0.967 to 0.406. Strengthening it ($b=4$) raises the correlation to 0.962 and 0.996. In TinyLlama all three quantities increase monotonically with b . In Qwen they peak at $b=2$ and fall slightly at $b=4$. TOHA’s own AUC barely moves (0.65 to 0.67 and 0.65 to 0.70). Its gap to the sink-score version is not monotone in b , so the intervention supports the mechanism at the level of the attention graph, not a dose-response in detection accuracy.

Table 13: **Sink-bias intervention** (teacher-forced TruthfulQA; labels unchanged). A constant b is added to the attention logit of key 0 for every query after token 0, in every head and layer. $\delta_P=0$: share of prompt-coned head graphs. Sink top: share of head graphs in which token 0 is every response token’s most-attended prompt token. ρ : median per-head rank correlation between TOHA’s d and the response sink score. Remaining columns: AUCs as in Table 4.

Model	b	$\delta_P=0$	sink top	$\rho(d, 1 - \bar{a}_0)$	TOHA	TOHA $[\bar{a}_0]$	TOHA $_{\text{sup}}$	$\bar{a}_{0,\text{sup}}$
Qwen2.5-1.5B	-2	37%	21%	0.605	0.667	0.624	0.843	0.849
Qwen2.5-1.5B	0	58%	45%	0.893	0.646	0.628	0.858	0.857
Qwen2.5-1.5B	+2	72%	60%	0.970	0.667	0.637	0.857	0.857
Qwen2.5-1.5B	+4	69%	56%	0.962	0.647	0.640	0.839	0.838
TinyLlama-1.1B	-2	63%	21%	0.406	0.698	0.626	0.845	0.834
TinyLlama-1.1B	0	74%	63%	0.967	0.666	0.618	0.845	0.848
TinyLlama-1.1B	+2	81%	75%	0.992	0.652	0.610	0.848	0.848
TinyLlama-1.1B	+4	86%	83%	0.996	0.665	0.616	0.844	0.842

G MISTRAL-7B

Mistral-7B-Instruct-v0.2 does not fit in the memory of the machine used for the main experiments. The released extractor runs it unchanged on a single 16 GB GPU (`sinktda/RUN_ON_5080.md`). An accuracy-level audit of this model under its [INST] format found that 99.99% of its 1D features are exactly zero and that 0D and 0D+1D banks are equivalent; by Theorem 1(c) this is what exact coning in every graph would produce, but we have not yet verified $\delta_0 = 0$ on its graphs directly.

H RELATION TO ACCURACY-LEVEL AUDITS, AND COST

Audits of attention-graph PH that stop at classification accuracy observe four facts that the theory now explains: (i) 0D persistence equals an MST statistic (Fact 1); (ii) 1D persistence adds little beyond 0D (Table 3, column “1D | 0D”), which Theorem 1 now explains; (iii) an earlier audit reported that the 1D bank of Mistral-7B under its [INST] format is 99.99% zeros, which is what exact coning would produce (not yet re-verified with this pipeline; Appendix G); (iv) HaluEval answers are separable from text alone.

Cost. Median per-matrix wall-clock time on real attention from TinyLlama-1.1B-Chat-v1.0 (sink features / δ_0 / dense MST / ripser 0D / ripser 0D+1D) is 0.001/0.02/0.10/0.02/0.02 ms at $N=32$; 0.001/0.03/0.21/0.03/0.03 ms at $N=64$; 0.001/0.07/0.42/0.07/0.06 ms at $N=128$; 0.001/0.25/0.90/0.18/0.18 ms at $N=256$. On i.i.d. random matrices of the same size, ripser 0D+1D takes 0.171 ms ($N=32$), 3.46 ms ($N=64$), 115 ms ($N=128$), 4,631 ms ($N=256$). On real attention the same computation is four orders of magnitude cheaper at $N=256$, consistent with Theorem 1: sink-dominated graphs have few, short 1D bars to compute. Sink features, read off one column, are one to two orders of magnitude cheaper still.